Physica A 468 (2017) 1–29

Contents lists available at ScienceDirect

Physica A

journal homepage: www.elsevier.com/locate/physa

Estimating pedestrian speed using aggregated literature data

Ernst Bosina *, Ulrich Weidmann

Institute for Transport Planning and Systems (IVT), ETH Zurich, Stefano-Franscini-Platz 5, 8093 Zurich, Switzerland

HIGHLIGHTS

- Walking speed data from more than 200 measurements were collected.
- A systematic description of influences on walking speed is provided.
- The most important influences on the walking speed were quantified.
- Standard conditions were proposed for better comparability of walking speed data.

ARTICLE INFO

Article history:
Received 21 August 2015
Received in revised form 21 August 2016
Available online 3 October 2016

Keywords:
Pedestrian
Walking speed

ABSTRACT

Since about 80 years, human walking speed has been measured for different purposes. The comparability of the results of these measurements is low, as the walking speed is influenced by a vast amount of parameters and no standardised measurement conditions exist. This work describes the most important factors influencing walking speed based on existing literature and estimates their impact. For this purpose, all walking speed measurements available in the literature were collected and reviewed. Using this data, the parameter values for different influences were computed and grouped according to their strength, which enables to determine the most important measurement parameters in future walking speed experiments. In addition, standardised measurement conditions are proposed which would provide a baseline for future studies and thus enhance their scientific value. For each of these conditions average walking speeds are calculated and the significance of the walking speed differences was determined.

1. Introduction

Engineers are interested in walking speed for the design of pedestrian facilities and the calculation of evacuation times. In medicine, walking speed is used as a measure for the health of human. Also in many other fields of research, walking speed is measured in combination with different influence factors. The walking speed of human has often been measured under various conditions and for different purposes. This variety of purposes produced many measurements describing a vast amount of factors influencing the walking speed. However, as most of the measurements are performed in different settings and only once, the possibilities for comparison are poor. In addition, depending on the specific measurements, their results differ substantially. However, in pedestrian simulations, the walking speed is an important input parameter, which strongly influences the results. Each of the influences has usually only a small impact on the results, but in combinations, they can generate considerable differences in the results, which are important especially for the design of pedestrian facilities. A better knowledge of the expected walking speed thus improves the reliability of the design.

* Corresponding author.
E-mail addresses: ernst.bosina@ivt.baug.ethz.ch (E. Bosina), weidmann@ivt.baug.ethz.ch (U. Weidmann).

http://dx.doi.org/10.1016/j.physa.2016.09.044
0378-4371/

In this study, the available data shall be consolidated to provide an overview of the available measurements. Here the influences and their correlations will also be systematically described. The next goal of this study is to extract best guesses based upon literature and to reduce the boundaries of parameters. By combining data from individual measurements, reasonable values can be provided. By analysing the data, the most important influence factors will be determined, too. Using the systematic description of the influences and the insights from the data analysis, potential research gaps, where measurement data is missing, can be identified. For future data collections, the paper also proposes standardised measurement conditions, which allow better comparability between measurements.

2. Method

2.1. Approach

The approach used in this work is based on a detailed data collection from literature. By systematically analysing existing literature, past walking speed measurement were identified and the results collected. This approach allows investigating measurement data from walking speed measurements in various settings and therefore having a broad overview about possible influences on the walking speed.

Compared to other approaches, this method has several advantages, but also drawbacks that have to be considered. First, the analysis of available data is an efficient method. No new experiments or real life measurements have to be done. To generate a similar amount of data already present in literature, a vast amount of resources have to be spent. It is therefore more resource efficient to first study existing data and try to get as much information as possible from this data. If this study reveals gaps in the available data, measurements that are more efficient can be done afterwards targeting these specific subjects.

On the other hand, the data available in literature is often lacking a detailed description of the precise conditions in which the measurements were performed. Statistical analyses and data comparisons are therefore strongly restricted. It also has to be noted that the study on influence factors often cannot be made under ceteris paribus conditions. Still, it is argued that the results of this study can produce benefit compared to existing data. At the moment, the values used for describing factors influencing the walking speed are either taken from single time measurements [1–3] or are also based on literature studies, but with a smaller data basis [4]. A study providing consolidated values from as much available data as possible therefore still provides added value to the existing knowledge on walking speed. Other methods, such as experiments or real life observations, also have shortcomings, which led to the decision to use a systematic literature review as the best suitable method. Apart from the resources needed for experiments, several influences, like the weather conditions, trip purpose or city size, cannot be tested in experiments with ceteris paribus conditions as well. In addition, pedestrians might react differently in the experimental context than under normal conditions. On the other hand, in real life observations it can never be assured that the set of pedestrians observed is comparable in different situations. Hence, a mixture of experiments and real life observations is needed to cover the whole range of potential influences, which leads to similar conditions as when using literature data. Especially cross correlations are complex to quantify, which is true for all possible methods.

2.2. Literature review

The literature review was done in a stepwise procedure. The first step was the determination of literature, which might include walking speed measurement data. For this, three approaches were used. By using literature already available in the author's own literature collection on pedestrian transport (based upon the references found in [4]), citations potentially including walking speed measurements were selected and the corresponding literature was collected. In addition, literature databases (e.g. Google, Google Scholar, Microsoft Academic, Scopus, ScienceDirect and TRID) as well as journal, publisher and author websites identified by the literature already found were searched for relevant citations. For the database searches, keywords like "speed", "walking", "pedestrian" and "human" in German as well as in English were used in different combinations. The third approach to find relevant literature was to look for citations including speed measurements within the available papers. Identified literature was then obtained both offline and online and then checked again for citations until no new useful documents were found. At the end of the first step, 573 out of 661 documents potentially including speed measurements could be obtained.

In the second step, the literature was evaluated and the walking speed measurements were extracted into a database. This was done in a chronological order starting with the first available walking speed measurements in 1933 [5]. This was done to ensure that the original sources were used and only new data was added to the database. In the end, 223 documents with unique measurement data were found.

As a third step, the influence factors considered in the following data extraction were determined. This was done by using the systematic developed by the authors in which they are categorised into four categories (Fig. 1). According to this scheme, the influence factors considered during data extraction can be distinguished into endogenous and exogenous as well as static and dynamic influences. Especially the classification of static and dynamic depends on the scope, here static is considered to change in time intervals larger than for example a walking trip or a short-term observation. Within these categories, the influences can be grouped into physical pedestrian properties, emotional and cultural influences, facility properties and environmental influences. For some of them, the detailed mechanisms linking them to the walking speed are missing, thus the classification needs to be further developed when these links are determined.

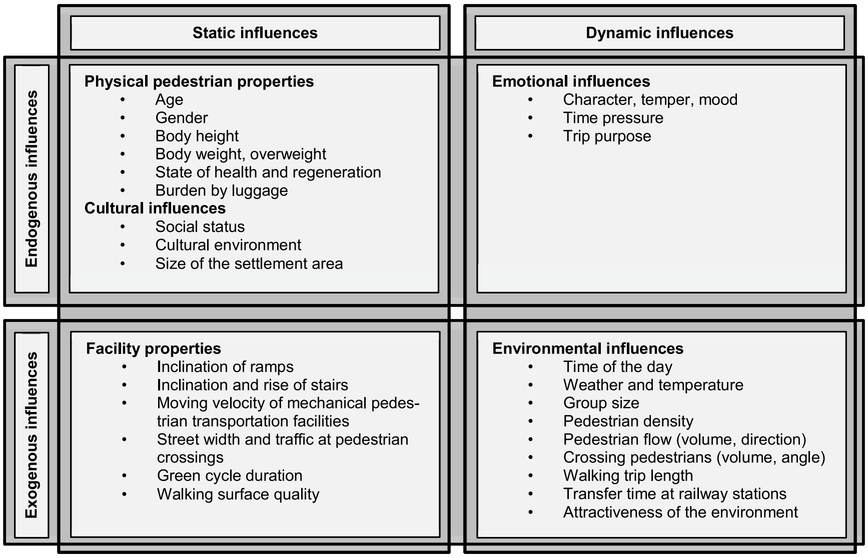

Fig. 1. Influences on the pedestrian walking speed, (own figure).

Table 1
List of influence factors used in the data analysis.

Chapter	Influence	Chapter	Influence
4	Physical pedestrian properties	7	Environmental influences
4.1	Gender	7.1	Group size
4.2	Age	7.2	Indoor/outdoor
4.3	Body height	7.3	Temperature
4.4	Body weight	7.4	Weather condition
4.5	Luggage	7.5	Year
5	Emotional and cultural influences	7.6	Month
5.1	Race	7.7	Day
5.2	Land use	7.8	Rush-hour
5.3	City size	7.9	Daytime
5.4	Country	7.10	Trip purpose
5.5	Continent	8	Observation technique
6	Facility properties	8.1	Measurement type
6.1	Inclination	8.2	Measurement method
6.2	Facility	8.3	Measurement length

In addition, the method used for the walking speed measurement and other parameters related to the observation as well as the aim of the measurement were considered as influence factors for the walking speed stated in the literature. These factors do not affect the walking speed per se, but they can influence the results of the measurements. Subsequently, a list of influence factors was created, which was used during data collection in the next step (Table 1).

As a fourth step, the data has been extracted from the literature. Based on the list of influence factors the corresponding values were determined for each measurement. Depending on the specific source, values for single measurements or mean values for a set of observations were available. It has to be noted that in most cases only a few influences were described for each study, hence the specific situation in which the measurements took place cannot be reproduced completely.

2.3. Data analysis

During data extraction, the first data analysis step was performed. Here, all literature was excluded which did not satisfy the requirements for data quality or comparability with other measurements. The most common criteria for exclusion were:

- The walking speed stated did not reflect the walking speed measured, but was for example already the result of a regression analysis.
- Only influences on walking speed were analysed which were not included in this study, for example different crossing types or the green cycle duration.
- The measurement method was disputable; hence, the quality of the measurement could not have been assured.

One of the main influences on the walking speed is the pedestrian density as walking speed declines with increasing density. To eliminate this strong influence, only walking speed measurements at densities where pedestrians usually walk unimpeded were used for this study. In free flow or similar conditions, pedestrians do not react to the presence of other

Table 2
Possible correlations between influences. Legend: ++ strong correlation, + correlation, ○ weak correlation, - no correlation expected.

	Gender	Age	Body height	Body weight	Luggage	Race	Land use	City size	Country	Continent	Inclination	Facility	Group size	Indoor/outdoor	Temperature	Weather condition	Year	Month	Day	Rush-hour	Daytime	Trip purpose	Measurement type	Measurement method	Measurement length
Gender																									
Age	+																								
Body height	+	++																							
Body weight	+	++	++																						
Luggage	○	-	-	-																					
Race	○	○	○	○	-																				
Land use	-	-	-	-	-	-																			
City size	-	-	-	-	-	-	++																		
Country	+	+	○	+	-	++	○	+																	
Continent	+	+	○	+	-	++	○	+	++																
Inclination	-	-	-	-	-	-	-	-	-	-															
Facility	-	-	-	-	-	-	+	+	+	+	+														
Group size	○	○	-	-	-	-	+	-	-	-	-	-													
Indoor/outdoor	-	○	-	-	-	-	++	-	-	-	++	++	-												
Temperature	-	-	-	-	-	○	-	-	+	+	-	-	-	++											
Weather condition	-	-	-	-	-	+	-	-	+	+	-	-	-	++	++										
Year	-	+	○	○	-	○	○	○	-	-	-	-	-	-	-	○									
Month	-	-	-	-	-	-	-	-	-	-	-	-	-	-	+	+	-								
Day	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	○	-	-							
Rush-hour	+	++	-	-	○	-	-	+	-	-	-	-	++	-	-	-	-	○	○						
Daytime	○	+	-	-	○	-	-	-	-	-	-	-	++	-	+	+	-	-	-	++					
Trip purpose	+	++	-	-	++	-	++	+	-	-	-	○	++	+	-	○	○	○	○	++	++				
Measurement type	+	+	-	-	++	-	++	-	-	-	-	++	++	++	+	+	-	-	-	++	-	++			
Measurement method	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	++		
Measurement length	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	++	++	

pedestrians, but freely choose their walking speed. This is often also reflected in the pedestrian fundamental diagram, where at low densities the walking speed is considered to be constant and only decreasing after a certain threshold, for example in (1993). Also in experiments with single file movement it was found that the velocity is constant up to a certain spatial headway [6]. Still, even at low speeds pedestrians react to other pedestrians, especially because not all pedestrians walk at the same speed. However, as pedestrians do not walk in lines, they can also move sideways to pass other pedestrians. Hence, they do not have to reduce their walking speed for overtaking other pedestrians as long as the density is low enough. For this study, we therefore define the free-flow speed as the speed where the pedestrian density does not influence the walking speed significantly. Based upon the data used for this study and theoretical considerations a pedestrian density of 0.5 P/m^2 was therefore used as the threshold value.

Next, the data points were aggregated to one average value for each situation with constant influence values. By applying this method, the differences in the aggregation level of the literature data as well as the influence of the different number of data points for a specific situation were eliminated. In general, more measurements will provide more reliable data, but this increase is nonlinear. As the amount of measurement values ranges from just a few to several thousand measurements, only the average value for each situation is used. This is done, as it is expected that the influence of different settings is more important, in terms of reliability of the results, than the amount of data collected in the same one.

For the data evaluation, each influence was analysed separately. On the one hand, this allows to evaluate each of them in a different way, which is a necessity considering the diverse influences. On the other hand, this approach assumes that the influence factors are independent; hence, no cross-correlation or bias between them occurs. In Table 2 possible correlations between the influence factors were estimated. For example, males are expected to be on average taller than women are. It can be seen that several correlations exist which cannot be studied by the method used in this work. It will also not be possible to solve this issue completely by using other methods. As it might affect the results for some influences, it will be addressed in the further analysis. Still, it can be seen that most of the influence pairs can be considered uncorrelated. For these cases, the results are meaningful.

Table 3
Order of influences for the analysis.

Order	Influence	Chapter	Number of studies	Number of measurements
1	Gender	4.1	77	253
2	Group size	7.1	13	33
3	Age	4.2	67	665
4	Inclination	6.1	9	14
5	Facilities	6.2	202	844
6	Indoor/outdoor	7.2	195	777
7	Body height	4.3	23	24
8	Body weight	4.4	19	19
9	Luggage	4.5	23	37
10	Temperature	7.3	4	6
11	Weather condition	7.4	8	40
12	Year	7.5	202	663
13	Month	7.6	32	157
14	Day	7.7	13	64
15	Rush-hour	7.8	202	632
16	Daytime	7.9	21	96
17	Measurement type	8.1	202	616
18	Measurement method	8.2	202	615
19	Measurement length	8.3	76	311
20	Trip purpose	7.10	202	554
21	Race	5.1	202	533
22	Land use	5.2	202	533
23	City size	5.3	157	325
24	Country	5.4	189	189
25	Continent	5.5	191	191

In addition, this work assumes that the strength of the individual influences is independent of other influences, for example, tall women react similarly to temperature as small men do. For the comparison of measurements in different settings, this assumption is crucial. Otherwise, it will never be possible to compare different measurements or generalise them, as it is impossible to generate exactly the same conditions twice. However, further research is indicated to determine and quantify these dependencies. Up to our knowledge, this was not done yet in a comprehensive way.

For this work, the individual influences were considered to reflect a percentage increase or decrease in walking speed compared to a reference condition. For example, women walking speed is considered to be 92% of men's walking speed. The threshold for the significance was set at 0.05. The superposition of different influences will then be done by multiplying their effects.

In general, two different types of influences in relation to the available measurements can be determined. A few influences, such as gender and age of a person, vary within one data collection. Hence, they can be compared with each other in a setting, where all the other influences can be assumed to be constant. For example, to quantify the influence of the gender, the walking speed for men and women were compared in the same situation. Depending on the situation, the absolute value of the walking speed will be different when comparing with another situation, but it is assumed that the ratio between the walking speed for women and men will be constant. The quantification of the influence was therefore done by taking the ratio between the walking speed and the walking speed for a given reference scenario (e.g. men's walking speed).

Other influences, such as the observation technique, mainly differ between different studies, but usually do not change within a single study. Therefore, these influences can only be evaluated by using data from different measurements, which might have varying measurement conditions. Here, the absolute values are given, although the relative values are more meaningful and the level of the absolute values might be biased. Nevertheless, as the determination of a reference condition is not suitable for most of the influences, no reference conditions were introduced at this point and no normalisation was done. In the conclusions, a proposition to resolve this issue and to enhance comparability between different measurements can be found.

At first, influences were analysed where a comparison with otherwise constant conditions was possible. Successively the quantity of each influence was estimated and then the influence was eliminated from the dataset. This was done by calculating the corresponding walking speed for a reference scenario, where the examined influence has a fixed value.

Afterwards, the other influences were ordered to first evaluate the ones, which are considered strongly independent from the others. For example, the facility type is considered to have an impact on the walking speed independent from the country of the measurement. An uneven distribution of the measurements concerning the country is therefore not expected to affect the influence of the facility type. Otherwise, the walking speed in a specific country will be strongly influenced by the facilities on which the walking speed measurements are performed. Therefore, the facility type will be evaluated prior to the country. The order of influences used for the evaluation can be found in Table 3. It has to be noted that in some cases, the order can affect the results of the analysis, so that the effect of latter influences is reduced in favour of earlier ones. For example, the influence of body height might be already included in the gender, as males are on average taller than women. This issue will be further discussed in the respective sections.

Table 4
Different groups of walking speed measurements.

Topic/goal	Main statement	First studies	Recent studies	No of studies used in the analysis
Social status, personal motivation	Successful and busy people walk faster	Jahoda et al. [5]	Wiseman [13], Morgenroth [14]	18
Capacity of pedestrian facilities	Creation of fundamental diagrams, maximum capacity	1937 in Ref. [15], Reimer [16]	Nazir et al. [17]	67
Pedestrian crossing	Pedestrians crossing speed, design speed	Evans [18]; Kirsch [19]	Hediyeh et al. [20]	42
Building evacuation	Walking speed for people in evacuation situations, capacity of bottlenecks	National Bureau of Standards [21]	Almejmaj and Meacham [22]	8
Modelling pedestrians	Human walking in predefined conditions for model evaluation	Daamen and Hoogendoorn [23]	Daamen et al. [24]; Nikolić et al. [25]	23
Medicine/human gait	Determination of the optimal walking speed in terms of energy consumption	Scholz [26]	Ludlow and Weyand [27]; Weyand et al. [28]	16
Medicine/health	Walking speed as an indicator of health	Browning et al. [29]	Studenski et al. [30]	15

For comparison, data from the literature study made by Weidmann [4] and translated to English [7] is used. This study on the walking speed was done more than 20 years ago using a considerable smaller database, but is still widely used as a reference for walking speed data [8–11]. Therefore, the results of this study are compared to the existing values and the differences are discussed.

3. Data description

3.1. Literature groups

The literature on walking speed measurements can be divided into different groups according to their research focus [12]. In addition, the grouping of the literature available revealed further research focuses. The main groups researching walking speed can be found in Table 4.

The first measurements found were done in a sociographic study about Marienthal, a small city with a high rate of unemployment during the World Economic Crisis [5]. Here the walking speed was used as a measure for the social status and their personal motivation. In the following decades several other publications were found, which correlate the walking speed with their cultural, economic and social status, formulated in the "pace of life" [2,31–33,14,34].

In the middle of the 20th century the first literature concerning the capacity of pedestrian facilities were found [35,18,16,26]. Until now more and more measurements under different conditions were performed. Some literature concentrates on the capacity of pedestrian facilities, hence strongly considering the speed–density relation [36,37,3,38,39]. A special focus is set here on the capacity and design of public transport stations [40,1].

In transportation research another aspect where the walking speed is required are pedestrian crossings. For the evaluation of the capacity and the design of traffic light cycle durations the expected walking speed is needed. Several literature is published investigating the walking speed for different crossing types and pedestrian characteristics [41,42,19].

Another engineering application for walking speed data is the building evacuation. Here often the walking speed on stairs is measured to determine the egress time [43,21,44,45].

In the last decades several approaches for modelling pedestrian traffic were proposed [46–49]. To validate these models, data on pedestrian walking speed under defined conditions is needed [50,51]. Therefore several experiments were performed, also to evaluate human walking in different geometries, such as turns and bottlenecks [52–54].

Parallel to the walking speed measurements done in the field of engineering, walking speed was investigated in medicine. First, the research focus was set at the understanding of human gait and the corresponding energy consumption [55–58]. Secondly, the walking speed was found to be a good indicator for the health of a subject. Therefore various literature was found investigating the influence of physical and mental disabilities to the walking speed [29,30].

In all of these fields, also studies from recent years can be found, indicating that there is still research conducted in this field and further walking speed measurements are performed. In Table 4 relevant studies from recent years are mentioned as well as the number of studies per field used in the further analysis. It has to be mentioned, that this number do not represent the total amount of literature in this field, as studies not relevant for further analysis are not stated.

3.2. Number of measurements

Most of the data obtained from literature is provided in an aggregated form. Whenever possible, the number of single speed measurements was recorded. When there was no information about the number available, each data point was

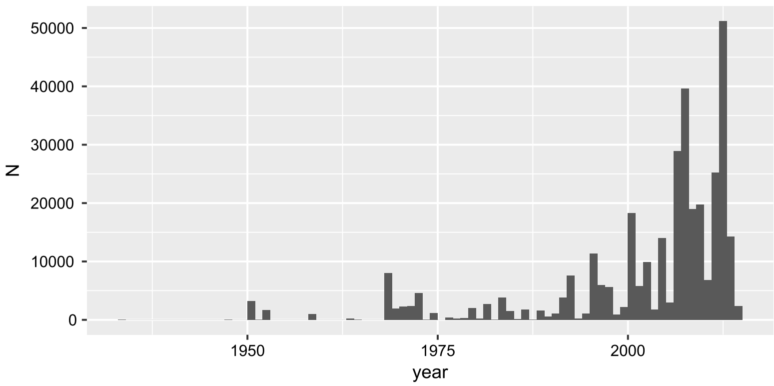

Fig. 2. Number of measurements obtained by year of publication.

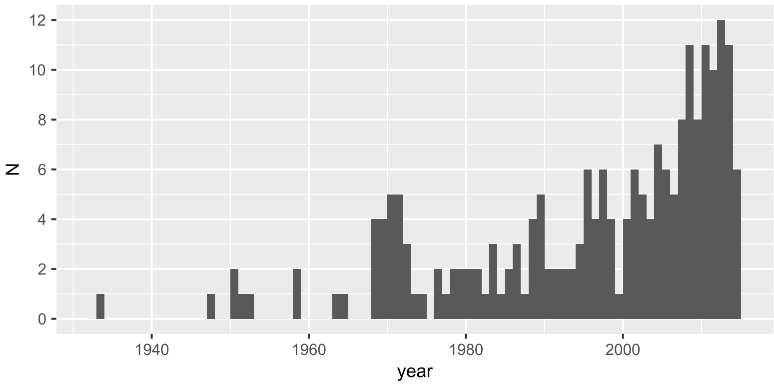

Fig. 3. Number of literature sources by year of publication.

recorded as one measurement. In total 342,576 measurements were retrieved from literature. Due to the broad literature review, it is expected that a considerable share of the total available data was obtained. Still, it has to be mentioned that measurements not published to a wider audience and sources in languages other than English and German might be missing.

Due to the possibility to perform long time video recordings and, in the last years, to extract speed data using automatic procedures, the amount of data increased within the last years (Fig. 2). As can be seen in Fig. 3, also the number of literature sources per year increased in the last decade, but not proportional to the amount of data. Thus, the number of measurements per literature source increased.

3.3. General statistics

The histogram of the speed measurements can be found in Fig. 4. It can be seen that the walking speed for free flowing conditions varies between 1.0 and 1.6 m/s with some outliers above and below. The peaks in the histogram can be due to several reasons. For some data, only the mean value for a certain set of influences is provided, hence the distribution of the values is neglected. Also for some influence factors, some values might be overrepresented. For example, walking speed of only elderly people is often determined to obtain the walking speed reduction when getting older.

One of the main influences to walking speed is the pedestrian density, as described in the fundamental diagram. For all the data, where the density could be obtained, Fig. 5 (all data without stairs) and Fig. 6 (data from stairs) show the relationship between speed and density. Where free flowing conditions were stated in the literature, a density of 0 P/m^2 was used. At low densities, the observed speed varies greatly. With an increase in density, the range of speed values observed reduces, mainly by the reduction of the maximum speed. Even at densities higher than 5 P/m^2, movement was measured. However, it has to be considered that only a few measurements are available for these high densities.

For comparison, the fundamental diagram curve from Ref. [4] for flat walkway respectively up- and downhill walking on stairs is added to the figures. These curves were obtained from a literature review and were calculated as the fundamental diagram for an average population. For flat surfaces, the curve is in a good fit at the upper part of the data, compared to the experimental data. For stairs, especially at low densities higher walking speeds were measured compared to the Kladek Formula. Considering the histogram for walking speed on stairs in Fig. 4 it can be seen that the range of walking speed measurements is higher than expected, but the average is in well accordance to the curve from Weidmann.

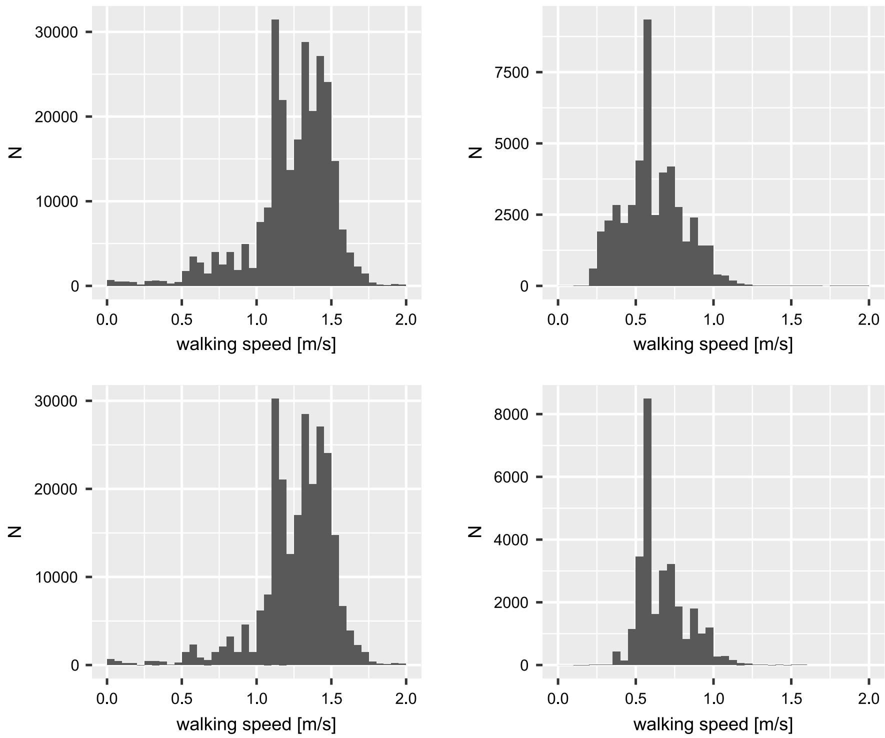

Fig. 4. Histogram of all speed measurements obtained from literature (left = non-stair facilities, right = stairs, top = all densities, bottom = densities ≤ 0.5 P/m^2).

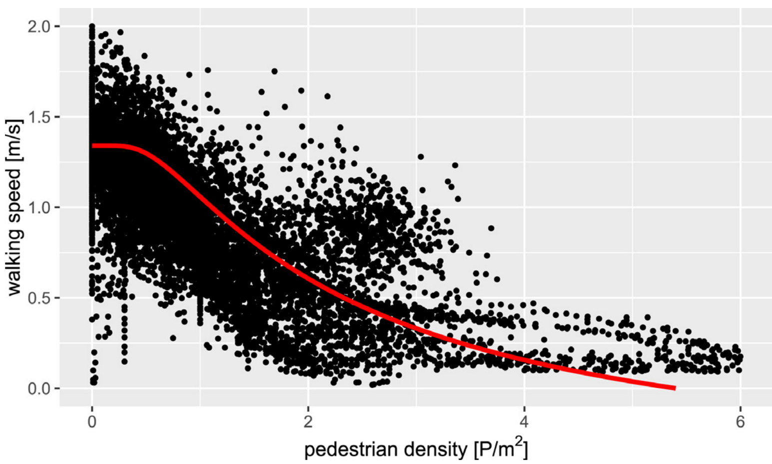

Fig. 5. Speed–density relation for all literature data except data from stairs used compared to the Kladek Formula [4].

4. Physical pedestrian properties

4.1. Gender

To compare the influences of gender on the walking speed, data from observations were used, where only the gender differed. Using this method, 117 data pairs from 65 literature documents were used. For this data, the ratio of women walking speed to men walking speed was calculated. When plotting this ratio against the men walking speed, a linear correlation can be estimated, which indicates that the ratio is equal for all walking speeds (Fig. 7). Therefore, no evidence was found that the difference in walking speed due to gender is affected by the density or other factors influencing the walking speed. The ratio between men's and women's walking speed ranges from 0.40 to 1.24 with an average of 0.92 (Fig. 8), which means that in average the women's walking speed is 92% of the men's walking speed. This is in good accordance to the value of 90% found by Weidmann [4]. The difference between men's and women's walking speed was found to be significant.

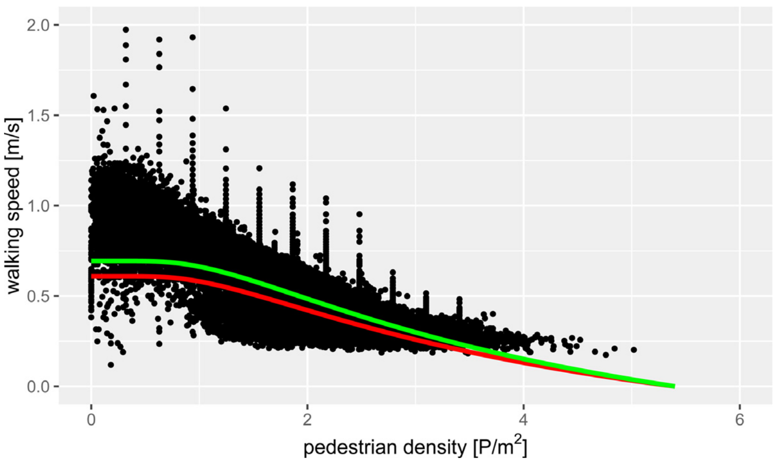

Fig. 6. Speed–density relation for all literature data for walking speed on stairs used compared to the Kladek Formula for walking up- and downhill on stairs [4].

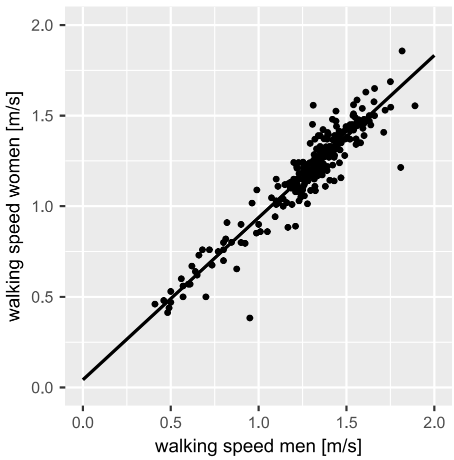

Fig. 7. Correlation between walking speed of men and women in same situations.

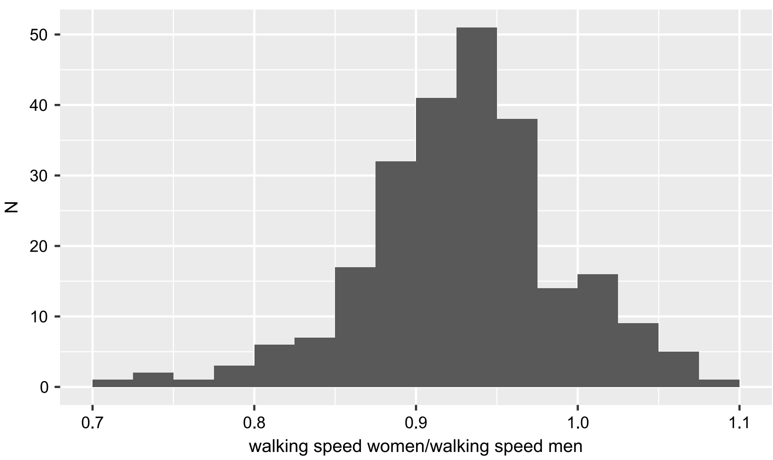

Fig. 8. Histogram of the ratio between the walking speed of woman and men.

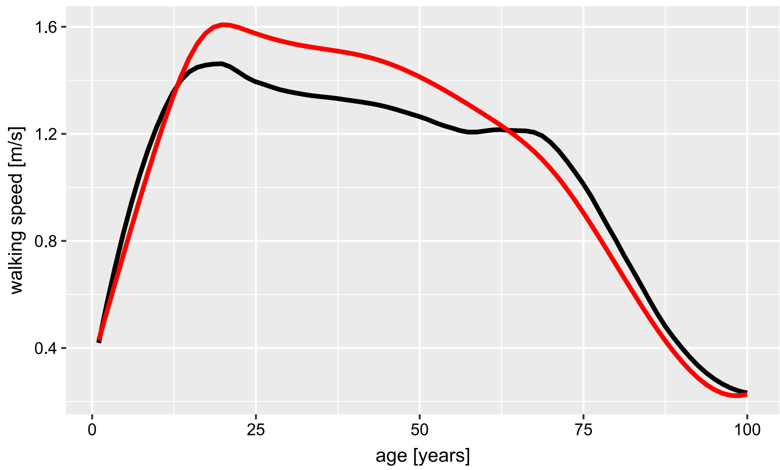

Fig. 9. Influence of age on the walking speed (black: new data, red: [4]). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.2. Age

Several references were found which clearly indicate the influence of age on the walking speed. The walking speed is considered to increase until the age of about 20 and then to decrease similar to the physical abilities [4].

To evaluate the influence of age based on the literature data, a reference curve obtained from Ref. [4] was used. This step was necessary for comparing speed values for different age classes. Based on the age class of each measurement, the corresponding speed value was obtained using the curve. To include the fact that the number of people in each age class is not the same, especially when considering elderly people, an estimate of the age distribution was used [59]. As most of the literature used was from Europe and North America, the age distribution for these continents served as a reference. The results were then compared to the literature value to obtain the ratio between the literature value and the estimate based on the speed-age curve. To eliminate other influences, the average ratio for each set of measurements was computed and used to normalise the measurements. As a result, the deviation of the literature measurement to the initial curve was obtained. The mean deviation for all literature values can be found in Fig. 9. It can be seen that the differences in walking speed between the age of 20 and 70 are found to be lower than in the reference data.

Most of the measurements were done either by estimating the age group of the pedestrian in field studies or by doing laboratory experiments. In these cases, people unable to walk are usually not considered. This group is relatively small in the younger age groups, but the group of old people include a higher amount of these people. In field measurements, also people are not measured who stay at home, because they are not able to cover longer distances or are too slow to cross the street at a reasonable amount of time. Therefore, it has to be considered that the speed values measured in the older age group do not represent the whole population.

4.3. Body height

Higher body height and therefore higher leg length is considered to be correlated to higher walking speed [4]. In the literature data used for this study, only in some references the body height was stated. Among these, no source was found where the velocity was measured with the body height as the only varying variable, usually also the age, gender or the body weight differs at the same time. Using a stepwise linear regression, speed measurements performed in several Swiss cities showed an increase in walking speed with increasing body height [60]. Here, the body height was estimated using three height categories, hence no quantitative information about the actual body height-speed relation is available.

Contradicting the literature results, the data in Fig. 10 show a decreasing trend in the walking speed with height. A possible explanation for this result is the small amount of data and the fact that there are several other parameters influencing the walking speed. Another effect might be that the range of body height in this study is narrow, hence differences in the energy expenditure for walking, which might influence the walking speed, are small and that pedestrians do not necessarily walk with their energetically optimal walking speed. Walking can be considered as a pendulum movement, thus body height positively correlates with speed by an increased step length due to longer legs but is negatively correlated due to a reduced step frequency. For similar body heights, as present in the data, this effect almost eliminates the height effect. Still, for small people like children it is intuitive that the walking speed is reduced. It also has to be noted that the body height is strongly correlated with other influences such as gender and body weight, thus the reliability of this result is small. At least for the body height studied, it can be concluded that the influence of body height is statistically

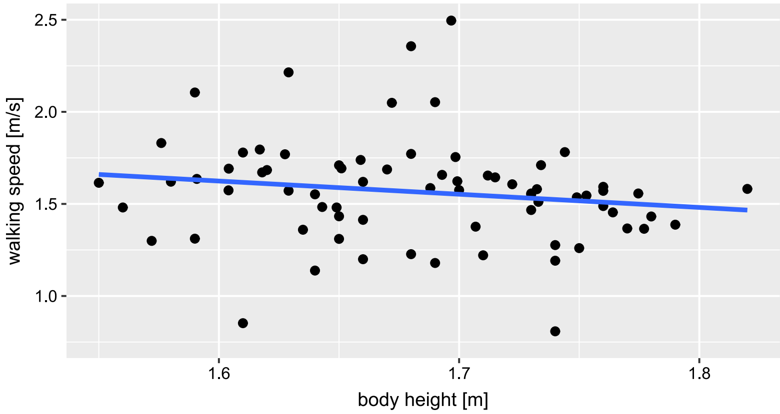

Fig. 10. Body height versus walking speed, normalised for age and gender ($y = -0.716x + 2.769$; $R^2 = 0.026$).

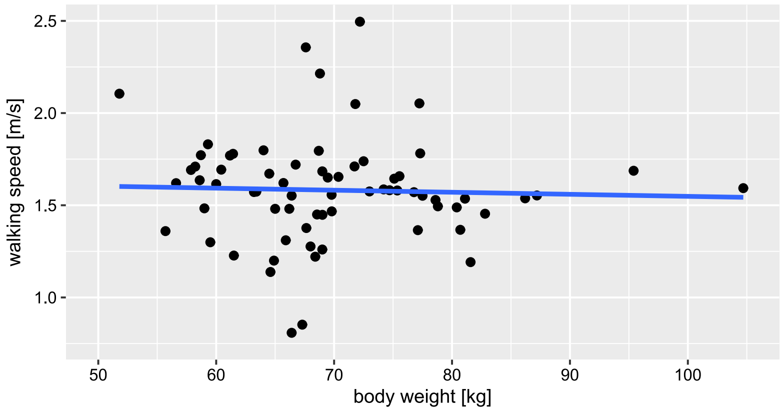

Fig. 11. Body weight versus walking speed ($y = -0.001x + 1.659$; $R^2 = 0.001$).

not significant, thus, if any, this influence is relatively small. As the amount of data available is small, the influence was not included in further analyses.

4.4. Body weight

Similar to the body height, the body weight was usually measured in combination with age or gender. Using a linear model, the results in Fig. 11 show a slight decrease in walking speed with increasing weight, although this decrease is not significant. This is in accordance with some of the references [60] whereas other studies did not show any influence [29]. Measurements on the energy consumption for walking show that the optimal walking speed in terms of energy is slower for heavier people [29]. Still, the energy demand does not change strongly at slightly lower or higher walking speeds, therefore the influence of body weight on the walking speed can be smaller than it is reflected in the energy demand. It has to be added that the body weight also correlates with the body height. This correlation and its influence on the walking speed were not investigated in detail here.

4.5. Luggage

Another influence factor is the luggage carried by the person. 13 references were found, which provided the walking speed for people carrying different kinds of luggage. For comparison reasons, only people with no luggage were compared to people with different kinds of luggage. As the different sources use different definitions of people carrying luggage or do not provide any definition, no further investigations were done. Comparing the influence of luggage carriage in different sources, no clear influence can be determined. The walking speed of people carrying luggage varies between 76% and 111% of the walking speed of people carrying no luggage. As it is usually not expected that people carrying luggage walk faster than people with no luggage, the reason for this behaviour might be that the two groups differ in some other influence factor, which was not observed. On average, pedestrians carrying luggage were observed to walk at 95% of the speed of people walking without luggage. Still, this difference was found to be not significant.

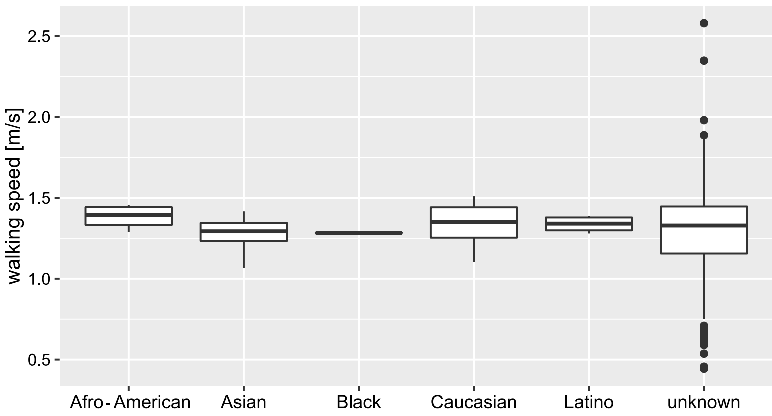

Fig. 12. Race versus walking speed ($R^2 = 0.001$).

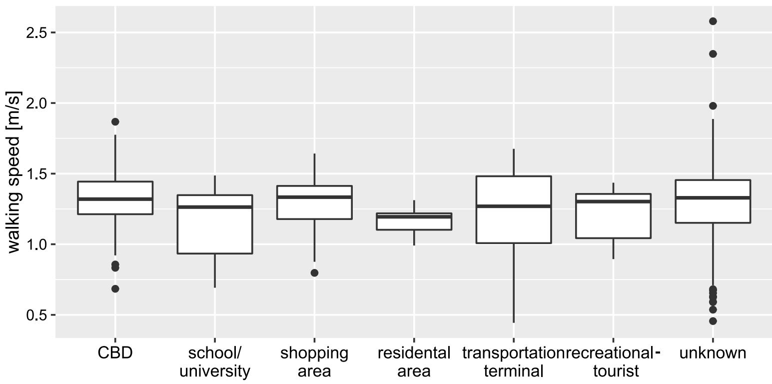

Fig. 13. Land use versus walking speed ($R^2 = 0.029$).

5. Emotional and cultural influences

5.1. Race

References were found indicating that the human race influences their walking speed [60]. As there is no reason, why the race itself has an influence on the walking speed and the differences are not statistically significant, the race might serve as an indicator for other influences (Fig. 12). For example, the personal characteristics or the trip purpose might be different for different groups. The differences between races observed in the data are therefore not considered relevant.

5.2. Land use

The influence of the land use can be due to two different factors. First, the surrounding land use can have a direct impact on the perception of the human and hence his walking speed. Second, the land use can serve as an indicator for the pedestrian's trip purpose, which is often not revealed in the studies. Central business districts therefore show the highest walking speeds (Fig. 13), but only the land use categories school/university has a significant different average walking speed.

5.3. City size

Several references found an influence of the city size on the pedestrian walking speed [31,61]. This influence is also visible in the available data, even though it is not significant and relatively small (Fig. 14). The influence of city size is expected to serve only as a proxy for other influences, such as land use or trip purpose. Therefore, the fact that no or almost no influence was found in the data is not unexpected, even though it contradicts the references found. For cities with about 10,000 inhabitants the average walking speed is computed to 1.33 m/s whereas for cities with 1,000,000 inhabitants it is 1.35 m/s. Thus, due to the small absolute differences, the influence of city size can usually be neglected.

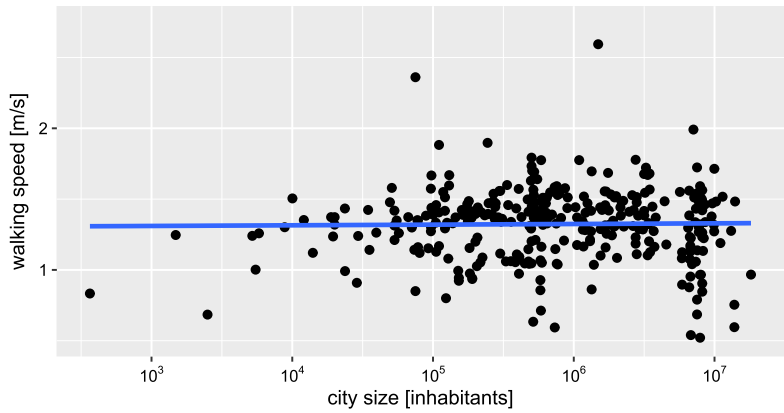

Fig. 14. City size versus walking speed ($y = 0.017 \log(x) + 1.264$; $R^2 = 0.003$).

Table 5
Average walking speed [m/s] in different countries (*based on own calculations*, $R^2 = 0.219$).

Australia	1.55	Kuwait	0.99
Austria	1.34	Malawi	0.59
Bahrain	1.06	Malaysia	1.42
Bangladesh	0.96	Malta	0.53
Brazil	1.38	Mexico	1.33
Bulgaria	1.25	Monaco	1.15
Canada	1.44	Netherlands	1.43
China	1.26	New Zealand	1.37
Costa Rica	1.36	Papua New Guinea	1.43
Croatia	1.54	Poland	1.48
Czech Republic	1.48	Romania	1.19
Denmark	1.73	Saudi Arabia	1.04
Egypt	1.32	Singapore	1.40
El Salvador	1.29	Slovenia	1.54
France	1.44	South Africa	1.26
GB	1.25	South Korea	1.32
Germany	1.37	Spain	1.59
Greece	1.27	Sri Lanka	1.37
Hungary	1.32	Sweden	1.33
India	1.16	Switzerland	1.32
Indonesia	1.16	Syria	1.20
Ireland	1.54	Taiwan	1.29
Israel	1.14	Thailand	1.05
Italy	1.33	United Arab Emirates	1.28
Japan	1.36	USA	1.33
Jordan	1.23	Yemen	1.31
Kenya	1.44	Zimbabwe	1.35

5.4. Country

Differences in the walking speed of different countries can have various reasons. As for most of the influence factors, they can occur due to other influences not stated in the observations and therefore not included. However, there are also real reasons, for example variations in climate, wealth, or culture. Usually the personal characteristics invisible for an observer are not known; therefore, it is impossible to explain the differences observed (Table 5). Even though some of the differences are significant, no conclusion can be drawn if these differences are linked to the properties of the country.

5.5. Continent

Summarising the countries by continent, some differences between continents can be found which are not significant. The continent with the slowest average walking speed is Asia, the fastest South America (Fig. 15). It has to be noted here that most of the measurements were done in Europe and North America, only a few measurements were obtained from Africa, South America and Australia.

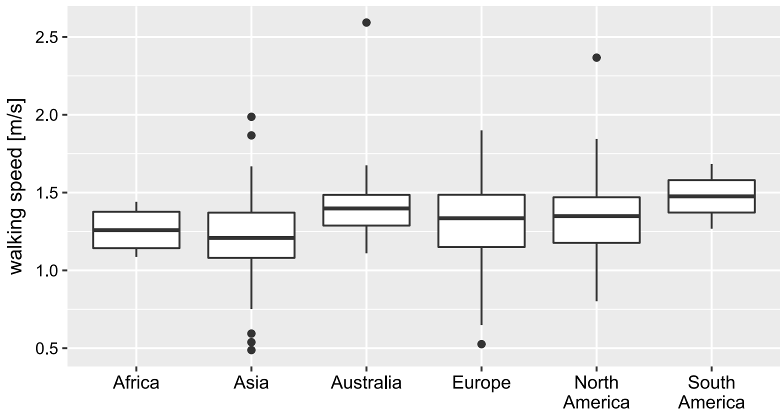

Fig. 15. Continent versus walking speed ($R^2 = 0.044$).

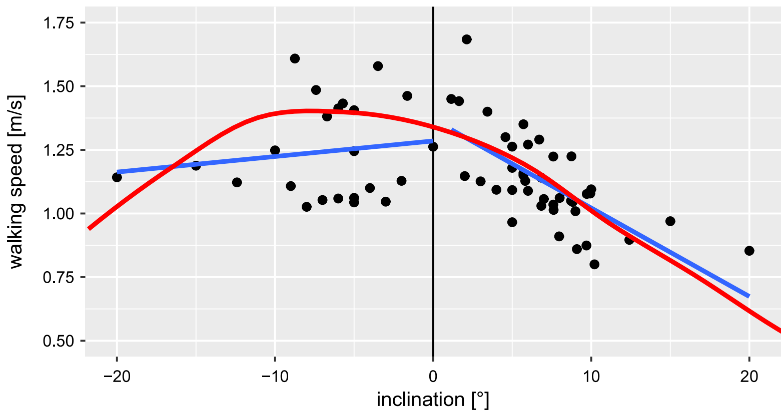

Fig. 16. Influence of the inclination [°] on the walking speed, (blue: new data, red: [4]) (*downhill* : $y = 0.007x + 1.289$; $R^2 = 0.021$, *uphill* : $y = -0.033x + 1.357$; $R^2 = 0.470$). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

6. Facility properties

6.1. Inclination

Several references include the effect of the inclination on the walking speed in their work. Based on the facility type, the inclination has to be divided into two different regimes. At higher inclinations, stairs can be found, whereas lower inclinations are usually connected to ramps. In Fig. 16 the walking speed in dependency of the inclination can be seen. Using the available data, a linear regression analysis was done separately for the up- downhill direction as well as for stairs and ramps. As expected all four analysis show an influence of the inclination on the walking speed, although only the influence of the uphill direction is significant. In the uphill direction on ramps, the reduction in walking speed was found to be highest. The data for the uphill direction is in well accordance with the reference values, whereas the downhill values were found to be lower. Here also the variation in the data is bigger, thus the results are less meaningful. A possible explanation for the only small influence in the downhill direction is that the power demand for walking downhill is lower than for flat walking, but the energy optimal walking speed and the willingness to increase the walking speed does not change much. Only for steep surfaces the walking speed will be reduced due to difficulties while walking, therefore this is not visible in the data.

6.2. Facility

The type of walking facility can be seen as one of the strongest influences on pedestrians walking speed (Fig. 17). Especially the presence of stairs leads to a considerable reduction in walking speed. Considering the limitations of the method used in this study, some of the differences in walking speed might also occur due to other influences. To increase the number of observations and therefore the stability of the results, the walking facilities were grouped by similar facility types Table 6. Surprisingly, the highest walking speed was found to be in malls. Here the small amount of data for this facility might have influenced the result. When using walkways as a reference, the categories stairs, pedestrian crossings, pedestrian zones and other show a significant different walking speed.

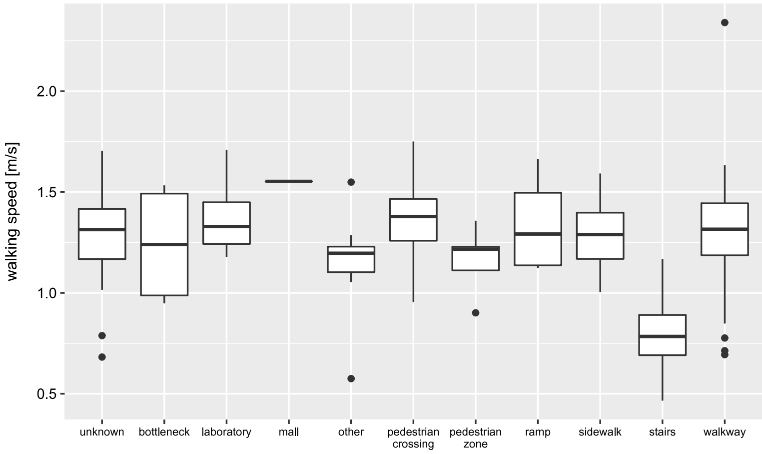

Fig. 17. Walking speed on different facilities ($R^2 = 0.342$).

Table 6
Average walking speed on different facilities.

Facility	Average speed
Bottleneck	1.24
Laboratory	1.36
Mall	1.55
Other	1.16
Pedestrian crossing	1.43
Pedestrian zone	1.20
Ramp	1.46
Sidewalk	1.32
Stairs	0.76
Walkway	1.34

7. Environmental influences

7.1. Group size

By using the same methodology as for the gender, the influence of the group size was determined. A group is defined as people walking together, hence having the same walking speed and adapt their behaviour to the others. This distinguishes them from people walking at higher densities, where they might also have the same walking speed and walking direction, but do not intend to stay in the vicinity of the other group members. In contrast 19 observations from 9 different sources were found, where the influence of the group size was determined. For comparison, the speed values for the different group sizes were put in relation to the speed at a group size of 2. In Fig. 18 it can be seen that an increase in group size decreases the walking speed. The highest reduction was observed between people walking alone and people walking in pairs, who have on average 90% of the single pedestrian walking speed (Table 7). The difference between the walking speeds for single pedestrian walking compared to groups of two to 6 pedestrians is significant. An increase in walking speed for groups of six people compared to a group size of five can be observed. Nevertheless, this difference is not relevant, as only a single source was found including measurements for groups of 6 people.

7.2. Indoor/outdoor

When comparing indoor and outdoor locations, a significant difference was observed. The data show that people tend to walk about 16% faster in outdoor areas. This observation can have several reasons, for example the influence of different air temperatures and other weather conditions, which correlates with the location of the infrastructure (outside or inside). Another reason might be different activities taking place in these environments. An obvious explanation of a direct connection between walking speed and the location of the study site was not found. For comparison, the data was normalised to outdoor conditions.

7.3. Temperature

Only few data was found with information given about the air temperature at the time of measurement. Still, this data show a significant trend to lower walking speeds in higher temperatures (Fig. 19), which is also in accordance with

Table 7
Average reduction in group walking speed.

Group size	Ratio between group walking speed and single pedestrian walking speed
1	1
2	0.90
3	0.84
4	0.79
5	0.77
6	0.78

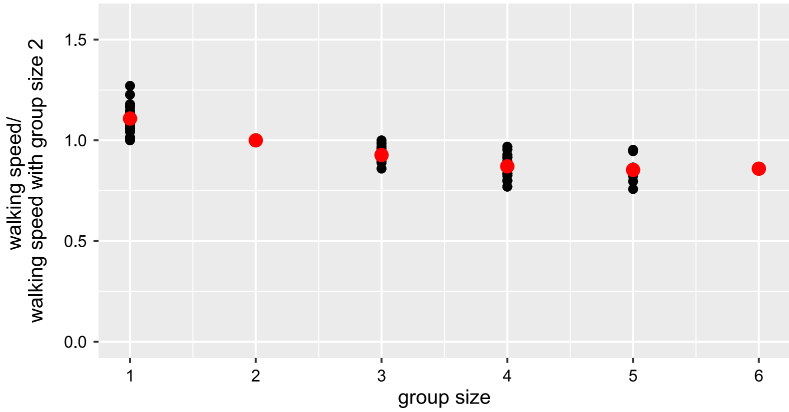

Fig. 18. Comparison on group walking speed to walking speed of groups of 2 people.

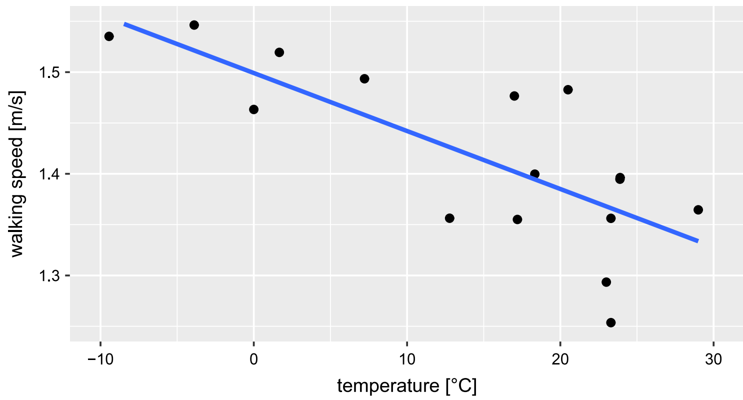

Fig. 19. Temperature versus walking speed ($y = -0.006x + 1.499$; $R^2 = 0.577$).

literature [62,63]. This behaviour seems reasonable, as in cold temperatures, people tend to avoid spending too much time outside and movement in areas with high temperatures is more exhausting. In addition, the temperature regulation of the human body supports this behaviour. Walking in areas with cold temperature helps to keep the body temperature high and therefore reduces the bodies energy demand for heating whereas in hot temperature areas walking increases the body temperature and thus increases the cooling demand.

7.4. Weather condition

Similar to the temperature, also other weather conditions might influence the walking speed (Fig. 20). Still, only a few references were found which do not allow making sound conclusions. As the differences are also not significant, the effect of the weather condition is not examined further.

7.5. Year

The year of measurement can be an indicator for several underlying influences. First, it is possible that the walking speed changed with time due to global trends not considered separately, for example the ageing society and the change in the social situation of the global population. Second, it might also show differences in the situations observed. As described

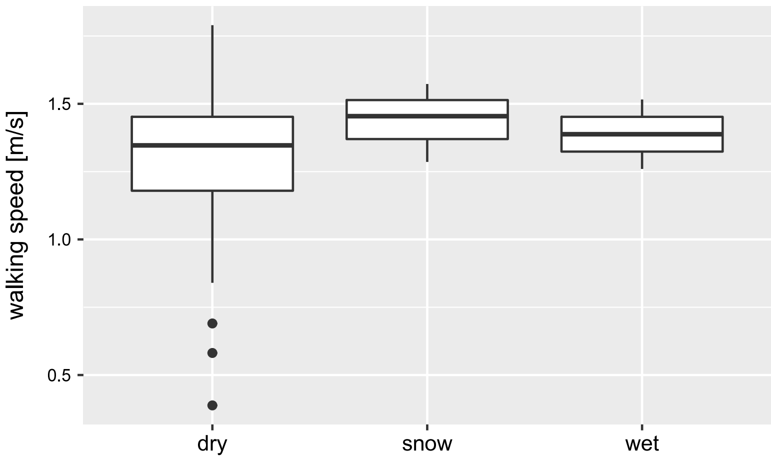

Fig. 20. Weather versus walking speed ($R^2 = 0.025$).

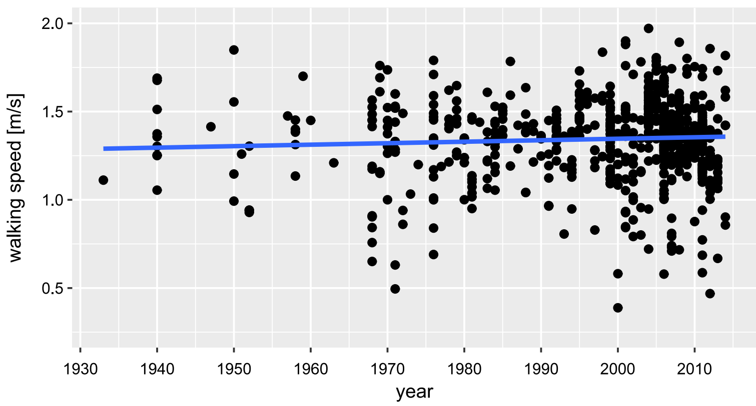

Fig. 21. Year of measurement versus walking speed ($y = 0.00060x + 0.1487$; $R^2 = 0.001$).

in chapter 3.1, the focus of the measurements shifted. A trend in the walking speed can therefore also show that not all parameters changing between these scopes are considered.

The analysis showed almost no trend (0.0006 $\mathrm{m/s} * \mathrm{year}$) which is also not significance (Fig. 21). It can thus be concluded that there is no influence from the measurement year to the walking speed. When considering the historic sources about walking distances, also no trend in the walking speed can be determined. In comparison between the Roman Empire and modern armies the maximum daily distance is about the same [64,65]. In former times, the walking speed was used in distance measurement units. The distance covered within one hour walking or within a day represents one walking hour or a daytrip. For one hour walking a distance of 3.7 to 5.7 km was assumed [66–71]. These values are in good accordance with the estimate of hourly walking distances used for hiking [72,73].

This fact also supports the outcome concerning the influence of the body height. Within the last century the average body height in Europe increased about one centimetre per decade [74]. As there is no visible trend within this time, this supports the earlier finding that the effect of human height can be considered negligible.

7.6. Month

The month of measurement can be used as an indicator for the weather. As the weather and temperature was not recorded for most of the datasets, it is expected that the influence caused by different weather conditions can still be observed comparing the different months. Still, the amount of data where the month of observation is stated is rather small. As most of the data was collected in Europe and North America, the temperature is highest between May and August and lowest in December to February. The analysis revealed that those months with warmer weather are related to a lower walking speed than the ones with cold weather, which is in good accordance with the expected outcome (Fig. 22). As there was not a high amount of date available, some months, especially March and April, show different results.

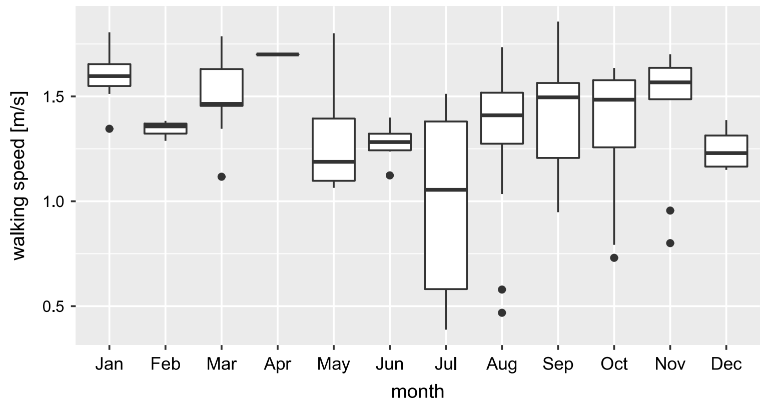

Fig. 22. Month versus walking speed ($R^2 = 0.193$).

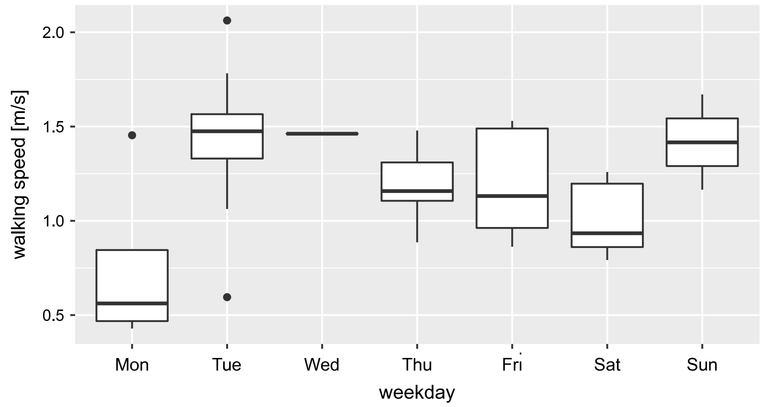

Fig. 23. Day versus walking speed ($R^2 = 0.399$).

7.7. Day

Surprisingly, the data indicates a considerable slower walking speed on Monday than on all other days (Fig. 23). The range of walking speeds measured for Monday indicates that other influences are also reflected in the data, which were not recorded and hence were not considered in the data analysis. In addition, the overall number of literature data stating the day of measurement is low and for specific days of the week, only few measurements are available.

When comparing weekdays with the weekend, the average walking speed during weekend was found to be within in the same range as the walking speed during weekdays. Still, Saturday shows the lowest walking speed excluding Monday.

There is no obvious reason why the day of the week as such influences the walking speed but there are other influences, like the trip purpose, which is different for the days of the week. For example, during the weekend less people are expected to walk to work. It is therefore usually expected that the walking speed on the weekend is lower. When measuring pedestrian speed, the trip purpose and other influences are often not determined. To compare short-term measurements, it would therefore be better to use similar days to eliminate the influences, which are combined with the day of the week. Analogue to mobility surveys, working days between Tuesday and Thursday should be used for walking speed measurements.

7.8. Rush-hour

Several references do not state the exact time, but indicate if the observations were made during rush hour or not. When comparing these two time types it can be seen that pedestrians tend to walk about 4% faster during rush hours (Fig. 24), but this difference is not significant. As an explanation for this, two different phenomena can be found. First, the physical capability of the human body changes during the day, having a maximum in the morning and the evening. Another factor is the differences in the set of pedestrians observed. Commuters consist of different age groups, but also have different trip purposes than the pedestrians observed outside rush hours.

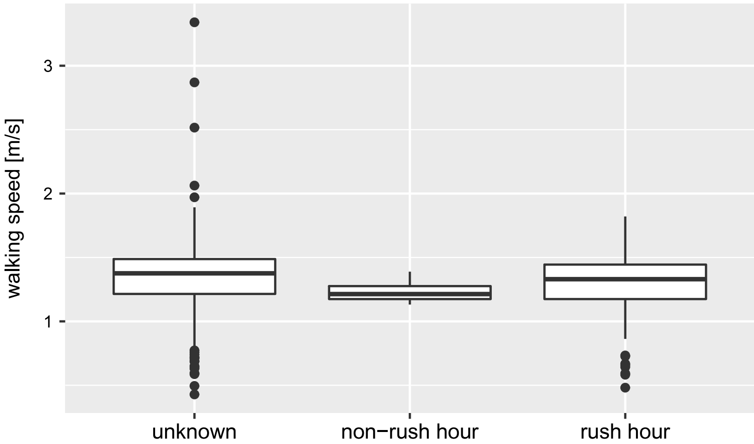

Fig. 24. Rush hour/non-rush hour versus walking speed ($R^2 = 0.009$).

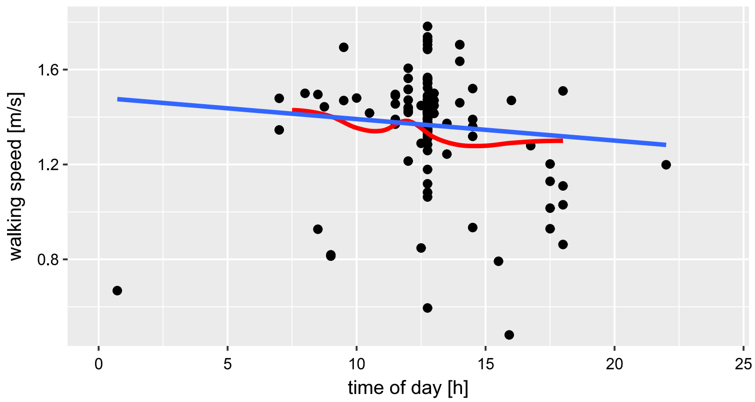

Fig. 25. Time versus walking speed compared with the curve from Ref. [4] ($y = -0.016x + 1.546$; $R^2 = 0.033$).

Table 8
Average walking speed for different trip purposes [4].

Trip purpose	Mean walking speed [m/s]
Commuting	1.49
Shopping	1.16
Business	1.61
Leisure	1.10
Average	1.34

7.9. Daytime

As already described in the previous chapter, several influences change with the time of the day. In Fig. 25, a decreasing trend in the walking speed throughout the day is visible. Nevertheless this trend is not significant. In addition, the scatter in the data is large, thus the influence of the linear trend is weak. A potential factor influencing the walking speed during the day is the physiological capability, which changes throughout the day [4]. In addition, influences, such as differences in the trip purpose or the age distribution can have an influence on the observed behaviour. For example, strong differences in the walking speed during the day between working and non-working days have been observed [75].

7.10. Trip purpose

Depending on the trip purpose, people walk with different speeds. As can be seen in Table 8, business and commuter traffic are considered to have the highest walking speeds, leisure trips the lowest [4].

The data obtained are in good accordance with this, but in general, the effects are smaller (Fig. 26). Only leisure trips show unexpected high walking speeds. Otherwise, the range of walking speeds in this group is high, which might indicate that the group consists of several subgroups or that other influences are present as well. When comparing with commuting, only

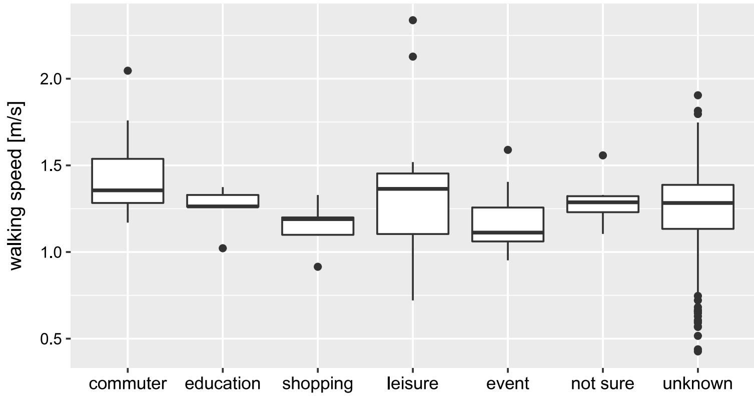

Fig. 26. Trip purpose versus walking speed ($R^2 = 0.032$).

Fig. 27. Measurement type versus walking speed ($R^2 = 0.009$).

the trip purposes shopping and event have a significant different mean walking speed. For further evaluation, the average of event, leisure, shopping, education and commuting was used as reference.

8. Observation technique

8.1. Measurement type

For data collection, different measurement settings can be used (Fig. 27). The most important ones are laboratory experiments, walking speed tests with single persons and real life observations. In comparison of these types, experiments have a slightly higher mean walking speed than real life observations (∼4%). Still, these differences are not significant. Some of this influence can be due to the different group compositions, which is partly already eliminated. In addition, the measurement setting itself can have an influence, as people might react different in real life and experimental settings.

8.2. Measurement method

Apart from the measurement setting also the method, which is used to obtain the speed measurements can have an influence. In general, the speed can be measured directly by personal taking the time a pedestrian walks a certain distance. Another method often applied is to record videos on site and extract the walking speed from these recordings. This can then be done either manually or using automatic extraction algorithms. As the speed of video recordings can be adjusted, manual walking speed measurements from video recordings are generally the most accurate, but also most time consuming, method. They are therefore used as a reference scenario. It is reasonable that the measurement method has an influence on the walking speed measured, as they all have different error sources, but the magnitude and direction is not clear. In Fig. 28 these differences can be seen, although only the mean walking speed of direct measurements and video recordings with automatic data extraction are significant different to video recordings with manual data extraction, which is used as reference.

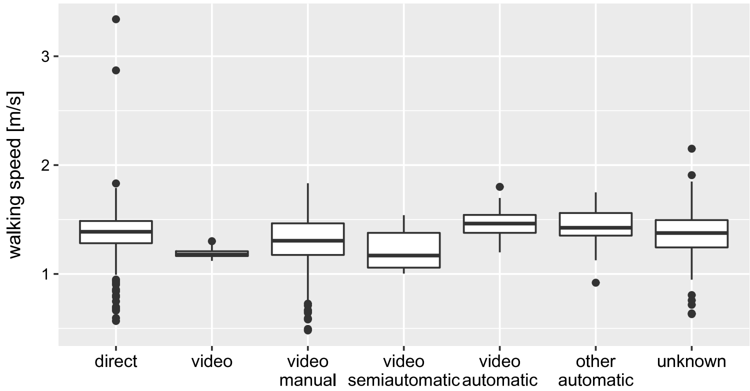

Fig. 28. Measurement method versus walking speed ($R^2 = 0.038$).

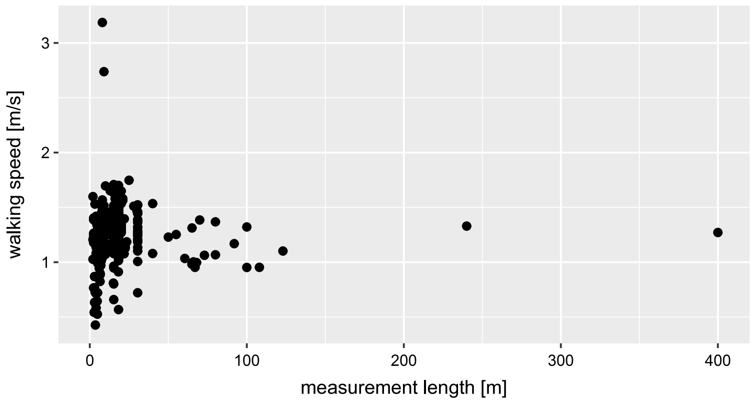

Fig. 29. Measurement length versus walking speed.

8.3. Measurement length

Field measurements were performed for short length up to a few hundred metres. It is expected that the length of the measurement section mainly has an influence on the variation of walking speeds, but not on the average value. Pedestrian walking speed is not constant, but can vary in short time intervals. When observing pedestrians on longer stretches the variation is therefore reduced. Still the mean value should be constant. As expected, the available data do not show any significant influence on the walking speed by the measurement length (Fig. 29).

9. Synthesis and conclusion

9.1. Strength of influences

The evaluation of the size of the different influences shows that they can be grouped according to their influence on the walking speed into four groups (A–D, Table 9). To describe the strength of the influences the range of mean values found for each influence was taken as a measurement. For example, 20-year-old pedestrians walk at an average speed of 1.6 m/s whereas people at the age of 90–100 years have an average walking speed of 0.2 m/s. Due to this big range of walking speeds age is classified in group A, having the strongest influence. These groups can be used to evaluate, which influences have to be considered, and hence described, when performing walking speed measurements.

The influences in group A have a strong impact on the result, thus their variation during the measurement period can have a stronger impact on the result than any other measurement, which should be evaluated. For one-time measurements, most of the influences in this group are usually constant; hence they have to be considered only when comparing different measurements. Otherwise, for example the age of the pedestrian is likely to be different when performing real life measurements. The most important influence on the walking speed, which always has to be considered, is the pedestrian density. Among all others, the age is the most important one, with differences between the fastest and the slowest group of up to 90% of the mean walking speed. Especially the occurrence of very young and very old pedestrians can have a strong impact on the total walking speed.

Table 9
Strength of influences.

Influence strength (maximum range of values in % of the average free flowing walking speed)	Influence	Number of studies used in the analysis
A (40%–100%)	Density	Not analysed
	Age	67
	Country	189
	Facility	202
	Inclination	9
	Day	13
	Daytime	21
	Month	32
B (10%–25%)	Group size	13
	Measurement method	202
	Trip purpose	202
	Continent	191
	Land use	202
	Indoor/outdoor	195
	Temperature	4
C (1%–10%)	City size	157
	Race	202
	Body weight	19
	Luggage	23
	Gender	77
	Measurement type	202
	Rush-hour	202
D (0%)	Body height	23
	Measurement length	76
	Weather condition	8
	Year	202

Table 10
Reliability of results.

Strong	Moderate	Weak
Gender	Body height	Luggage
Age	Body weight	Race
City size	Facility	Land use
Inclination	Indoor/outdoor	Country
Group size	Month	Continent
Temperature	Rush-hour	Weather condition
Year	Trip purpose	Day
	Measurement type	Daytime
	Measurement length	Measurement method

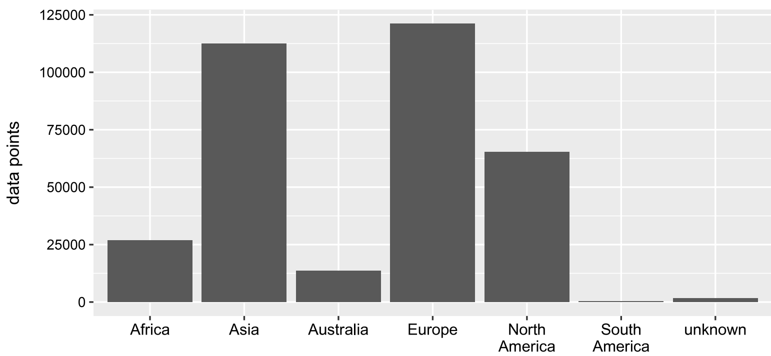

Fig. 30. Data points per continent.

Group B summarises influences which are still important to consider, but whose effect on the walking speed is much smaller. Influences in group C have only a small impact on the walking speed; hence it is not as important to record them. Still, when considerable changes are expected between measurements, their values should be recorded. Group D contains all influences, where no effect on the walking speed was found or their effect is considered negligible.

Table 11
Proposed standard conditions (**bold**: reference attributes for comparison at different locations).

Influence	Standard condition
Gender	**50:50** Comment: global average
Age	Age distribution, age group (1-, 5- or 10-year intervals) or: Young: 1–10 **Middle age: 10–70** Elderly people: 70+ Comment: Before 10 and after 70 years of age the speed decreases strongly.
Body weight & body height	For laboratory tests: state values All other: **random choice at location**
Luggage	Division between people having heavier luggage (pushchair, suitcase, heavy and big bags or backpack) and **people without luggage or small luggage (small bags/backpacks, handbags)** Comment: Influence of luggage is especially important at places with a high share of people with heavy luggage (airports, shopping centres etc.)
Race	**Random choice at location**
Land use	Description of land use at location: – **Central business district** – Recreational/touristic area – Residential area – School/university area – Shopping area – Transportation terminal
City size	**State value**
Country	**State country of measurement**
Continent	**Derived from country**
Facility—inclination	**Flat** Or state inclination and walking direction
Facility—type	Description of location – Stairs – Laboratory – Bottleneck – Pedestrian crossing – Pedestrian zone – Walkway—next to car traffic – **Walkway—separate from car traffic**
Group size	**Alone** Or for each group size separate value
Indoor/Outdoor	Description of location – **Indoor** – Outdoor
Temperature	State temperature (e.g. hourly average) **Standard**: **15–25** °C
Weather condition	State condition – Sun – **Cloudy** – Rain/precipitation – Snow
Time of measurement	State time of measurement: (year, month, day, time, rush-hour/non rush-hour) **Standard: Tuesday–Thursday, evening rush-hour (3–5 pm)**
Trip purpose	Estimation of main trip purpose – **Commuting** – Shopping – Business – Leisure – Average

(continued on next page)

Table 11 (*continued*)

Influence	Standard condition
Measurement type	State type – **Real life observation** – Walking test (single person walking) – Experiment
Measurement method	State method – Direct (manual timing) – Videotaping ∘ **Manual data extraction** ∘ Automatic data extraction ∘ Semiautomatic data extraction – Other automatic data extraction technologies
Measurement length	State length **Standard: 5 m** Comment: Even though, no influence from the measurement length was found for the mean walking speed, it is likely that the variation of the walking speed will be influenced, and thus a standard length shall be set for comparison reasons. When measuring walking speed in different densities, a measurement length of 5 m also reflects the distance, where people react to the pedestrian density present [76].
Pedestrian density	**Free flowing conditions** or state density in measurement area

The strength of the stated influences is not backed up very well in all cases. Depending on the amount of measurements available and their independence to others, there are several uncertainties occurring. In group A for example this is true especially for the influences country, day, daytime and month where other influences not considered might have a strong impact on the result.

9.2. Limitations

Throughout the study, several limitations and possibilities for bias in the results have to be considered. They can occur due to the data used in this study, the analysis of the data and the interpretation of the results.

9.2.1. Literature data

The data used in this study was obtained from literature available to the authors and in a language understandable for them. The availability of the literature restricts the data to data already published and available either online throughout inscriptions or freely available in the internet or as a hard copy available to the library system. The language restriction reduces the literature sources used to literature in English, German and some sources in other Indo-European languages.

As neither the measurements nor the literature are evenly distributed among all possible influence values, for some influences far more data is available. In addition, the amount of measurement values differs strongly between different sources. As an example, Fig. 30 shows the data points available per continent. It can be seen that most of the data was obtained in Asia and Europe and almost no data was available from South America.

The approach used in this study allows including data from various measurements done in the past to improve the knowledge about influences on the walking speed. Still, the use of data from literature has limitations to the results compared to data directly obtained from observations.

Depending on the specific source, the field surveys and laboratory experiments are described different; therefore often information irrelevant for the specific publication is not included. For this study, as detailed information as possible about the observation setting is important but missing. The differences between measurements can therefore also result from influence factors not stated in the literature.

The influence of the limitations related to the literature data to the results of the study is small compared to other limitations. The lack of data for all situations and the limitations in identifying suitable literature reduces the amount of data useable for the study and the parameters, which can be evaluated. As the amount of data is still relatively high compared to other studies, the influence of missing data is small. The main influence of this limitation is the higher uncertainty of the results, especially for influences with a limited amount of measurements. This leads to the fact that the deviation between measurements is still high, when all influences stated are brought to a reference scenario.

In this step, the main limitation is the missing description of influences in the study. This limits the comparability of different studies, as the speed differences observed might result from influences not stated in the studies. For future studies, it is therefore recommended to describe at least the states of the influences in group A and B according to Table 9.

9.2.2. Data analysis

An important limitation is the separate evaluation of each influence. This assumes independent parameters, which cannot be ensured in all circumstances. As described in the method section, the effect is taken into consideration for the evaluation

Table 12
Walking speed for the proposed standard conditions. A walking speed of 1.34 m/s is assumed for the reference attributes.

Influence (reference)	Attribute values (% of reference walking speed (1.34 m/s) at otherwise reference conditions [m/s])										Significance
Gender (50% men, 50% women)	men*	women*									significant
	104%	96%									
	1.39	1.29									
Age (10 - 70 years)	1 - 10	10 - 20	20 -30	30 - 40	40 - 50	50 -60	60 - 70	70 - 80	80 - 90	90 - 100	significance not determined
	66%	109%	106%	102%	99%	93%	92%	75%	43%	21%	
	0.88	1.46	1.41	1.36	1.32	1.24	1.24	1.00	0.58	0.29	
Body height	no influence found										no significant influence
Body weight (average)	50 - 60	60 - 70	70 - 80	80 - 90	90 - 100						not significant
	101%	101%	100%	99%	99%						
	1.36	1.35	1.35	1.34	1.32						
Luggage (no/small luggage)	no/small luggage	big luggage									not significant
	100%	97%									
	1.34	1.30									
Race	no influence found										no significant influence
Land use (central business district)	central business district	recreational/ touristic	residential area	school/university area*	shopping area	transportation terminal					some significant differences
	100%	92%	88%	88%	97%	91%					
	1.34	1.23	1.19	1.17	1.30	1.22					
City size (10.000 inhabitants)	10.000	100.000	1.000.000	10.000.000							not significant
	100%	100%	100%	101%							
	1.34	1.34	1.35	1.35							
Country	see Table 5										some significant differences
Continent (Europe)	Asia	Australia	Europe	North America	South America						not significant
	93%	111%	100%	103%	113%						
	1.25	1.49	1.34	1.38	1.51						
Facility – inclination (0%)	15% down	10% down	5% down	0%	5% up*	10% up*	15% up*				uphill: significant, downhill: not significant
	92%	95%	97%	100%	88%	75%	63%				
	1.24	1.27	1.31	1.34	1.17	1.01	0.84				
Facility – type (walkway – separate from car traffic)	stairs*	laboratory	bottleneck	pedestrian crossing*	pedestrian zone*	walkway – on street	walkway – separated				some significant differences
	57%	101%	93%	107%	90%	100%	100%				
	0.76	1.35	1.24	1.43	1.20	1.34	1.34				
Group size (1)	1	2*	3*	4*	5*	6*					significant
	100%	90%	84%	79%	77%	78%					
	1.34	1.21	1.12	1.05	1.03	1.04					
Indoor/outdoor (outdoor)	outdoor	indoor*									significant
	100%	86%									
	1.34	1.15									
Temperature (15 - 25°C)	-5 - 5°*	5 - 15°*	15 - 25°	25 - 35°*							significant
	108%	104%	100%	96%							
	1.45	1.40	1.34	1.28							
Weather condition (Dry)	dry (sun/cloudy)	rain/precipitation	snow								not significant
	100%	108%	112%								
	1.34	1.45	1.50								
Time of measurement: year (2000)	1950	1960	1970	1980	1990	2000	2010				not significant
	98%	98%	99%	99%	100%	100%	100%				
	1.31	1.31	1.32	1.33	1.33	1.34	1.35				

(continued on next page)

Table 12 (*continued*)

Time of measure-ment: month (Average)	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	significance not determined
	116 %	97%	110%	123%	93%	93%	71%	99%	103%	98%	107%	91%	
	1.55	1.30	1.47	1.65	1.24	1.25	0.95	1.33	1.38	1.31	1.43	1.21	

Time of measure-ment: weekday (Tuesday)	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday	significance not determined
	52%	100%	100%	81%	83%	69%	97%	
	0.69	1.34	1.35	1.09	1.11	0.93	1.30	

Time of measure-ment: daytime (3 - 5 pm)	7 - 9 am	9 - 11 am	11 am - 1 pm	1 - 3 pm	3 - 5 pm	3 - 7 pm	7 - 9 pm	not significant
	109%	107%	105%	102%	100%	98%	95%	
	1.47	1.44	1.40	1.37	1.34	1.31	1.28	

Time of measure-ment: rush-hour (rush-hour)	rush-hour	non rush-hour	not significant
	100%	97%	
	1.34	1.30	

Trip purpose (Commuting)	commuting	shopping*	leisure	education	event*	some significant differences
	100%	81%	96%	88%	83%	
	1.34	1.08	1.28	1.18	1.11	

Measurement type (real life observa-tion)	real life observation	walking test	experiment	not significant
	100%	103%	106%	
	1.34	1.38	1.42	

Measurement method (video: manual data extraction)	direct: man-ual timing*	videotaping	video: man-ual data ex-traction	video: auto-matic data extraction*	video: semi-automatic	other auto-matic data extraction	some significant differences
	105%	92%	100%	113%	95%	110%	
	1.40	1.23	1.34	1.51	1.27	1.47	

Measurement length	no influence found	no significant influence
Pedestrian density	not investigated	-

* Attributes significantly different to the reference conditions ($p < 0.05$).

order of the influence factors. For comparing single parameters, only this parameter should vary (ceteris paribus), whereas other influences are equal. This can only be guaranteed for a few situations, which makes this a major downside of this method. Still, as described previously, other potential methods will be affected by similar limitations.

As the data is often already aggregated, the variation of the walking speed cannot be evaluated using this method. In general, the variation in the data can be due to different influence factors or due to the variation within the same situation. For known influences, the variations of the mean value can be calculated with the proposed method. The effect of influences not described in the data and the variation in a specific situation cannot be evaluated here.

The limitation in the data analysis based on the data used for this study can strongly affect the results. To tackle these limitations, the influences analysed in this step were evaluated in a predefined order. This order aimed at ensuring that influences, which might have an impact on the results on the data analysis of other influences were examined before them. For example, the facility type was analysed before the continent as it was expected that different facilities influence the results for the continent stronger than vice versa. Here, also the amount of data for each influence was considered.

9.2.3. Interpretation

For the interpretation, the limitations of the validity of the results have to be considered. As described previously, the results might have a bias due to several assumptions and limitations made in this study. Especially when considering influences with smaller amounts of data available, the size of the influences is quite uncertain. In Table 10 an estimate was done to evaluate the reliability of the results of the influences considered in the study. It can be seen that for several influences the reliability is weak, thus further research is needed to quantify these influences.

9.3. Conclusion

It was possible to describe the influences on the walking speed based on literature data. The results were compared with data from previous studies, showing that most of the influences can be reproduced. Considering the insights gained from this study, the average walking speed of 1.34 m/s proposed by Weidmann [4] can be confirmed as a good estimation. Still, it has to be noted that providing an average walking speed is a difficult task, as for several influences (e.g. Land use, facility type, trip purpose, measurement method etc.) no average value and hence no average speed can be easily determined. This walking speed can thus be considered as an average of the walking speed measurements done in the past rather than a global average. It was further found that various influences have a relevant impact on the walking speed; hence, they should be considered in simulation and other applications. In addition, it was possible to reliably quantify several influence factors.

Still, it has to be noted that the differences of the mean walking speed between different measurements cannot be solely explained by the influences examined. One of the main reason for this are the different settings, which are used for the speed measurements and missing information in the literature concerning the detailed measurement setting. Therefore, several influence values are different between the measurements, but they are not described in literature. Hence, standard conditions are proposed in Table 11. Using this scheme, it will be possible to describe the measurements in a standardised way and thus enhance the comparability of different measurements. Information exceeding this scheme will further enhance studies on walking speed differences.

To further understand the underlying principles determining the walking speed, the correlations between different influences as well as their superposition has to be addressed. This can be done by applying multivariate statistics as well as by studying not only the influence factors but also the way they influence walking. For example, luggage increases the weight a person has to carry and hence the energy optimal walking velocity is shifted. If these connections are known for all the influences, correlations and dependencies can be better described. In addition, influences like country or continent can be determined, which only effect walking speed indirectly. For these, the direct influences can be identified.

Combining the data analysis done with the proposed standard conditions in Table 11, an estimate of reference values for the influence attributes can be given (Table 12). As a reference value, an average walking speed of 1.34 m/s was used. Using this table, it is possible to compare study results not measured using the same conditions. Future measurements using the standard condition scheme will ease comparability and further enhance the data basis for studying and understanding the pedestrian waking speed.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at http://dx.doi.org/10.1016/j.physa.2016.09.044.

References

[1] B.D. Hankin, R.A. Wright, Passenger flow in subways, Oper. Res. Q. 9 (2) (1958) 81–88. http://dx.doi.org/10.2307/3006732.
[2] Marc H. Bornstein, Helen G. Bornstein, The pace of life, Nature 259 (5544) (1976) 557–559. http://dx.doi.org/10.1038/259557a0.
[3] Francis P. Navin, Robert J. Wheeler, Pedestrian flow characteristics, Traffic Eng. 39 (4) (1969) 30–36.
[4] Ulrich Weidmann, Transporttechnik der Fussgänger - Transporttechnische Eigenschaften des Fussgängerverkehrs (Literaturauswertung). Zweite ergänzte Auflage. Schriftenreihe des IVT 90, Institut für Verkehrsplanung, Transporttechnik, Strassen- und Eisenbahnbau, ETH Zürich, Zürich, 1993, http://dx.doi.org/10.3929/ethz-a-000687810.
[5] Marie Jahoda, Paul F. Lazarsfeld, Hans Zeisel, Die Arbeitslosen von Marienthal, Suhrkamp Verlag, 1933.
[6] C. Appert-Rolland, A. Jelic, P. Degond, J. Fehrenbach, J. Hua, A. Cretual, R. Kulpa, et al., Experimental study of the following dynamics of pedestrians, in: Ulrich Weidmann, Uwe Kirsch, Michael Schreckenberg (Eds.), Pedestrian and Evacuation Dynamics 2012, Springer International Publishing, 2014, pp. 305–315. http://dx.doi.org/10.1007/978-3-319-02447-9_25.
[7] Stefan Buchmüller, Ulrich Weidmann, Parameters of Pedestrians, Pedestrian Traffic and Walking Facilities. Schriftenreihe des IVT 132, Institut für Verkehrsplanung und Transportsysteme (IVT), ETH Zurich, Zürich, 2006.
[8] Fiammetta Venuti, Luca Bruno, An interpretative model of the Pedestrian fundamental relation, C.R. Méec. 335 (4) (2007) 194–200. http://dx.doi.org/10.1016/j.crme.2007.03.008.
[9] Ronald John Galiza, Luis Ferreira, Phil Charles, Estimating the effects of older people in pedestrian flow: a micro-simulation approach, in: Transportation Research Board 90th Annual Meeting, 2011.
[10] Anders Johansson, Tobias Kretz, Applied Pedestrian modeling, in: Alison J. Heppenstall, Andrew T. Crooks, Linda M. See, Michael Batty (Eds.), Agent-Based Models of Geographical Systems, Springer, Netherlands, 2012, pp. 451–462. http://dx.doi.org/10.1007/978-90-481-8927-4_21.
[11] Winnie Daamen, Serge Hoogendoorn, Calibration of Pedestrian simulation model for emergency doors by Pedestrian type, Transp. Res. Rec.: J. Transp. Res. Board 2316 (2012) 69–75. http://dx.doi.org/10.3141/2316-08.
[12] Ulrich Weidmann, Der Fussgängerfluss in der Verkehrsforschung - Rückstand und Dynamik. Wege und Geschichte, 2012.
[13] Richard. Wiseman, Quirkologie: die wissenschaftliche Erforschung unseres Alltags, Fischer Taschenbuch Verlag, Frankfurt am Main, 2008.
[14] Olaf Morgenroth, Zeit und Handeln: Psychologie der Zeitbewältigung, Verlag W. Kohlhammer, Stuttgart, 2008.
[15] Wsevolod Michailowitsch Predtečenskij, Anatoli Iwanowitsch Milinskij, Personenströme in Gebäuden: Berechnungsmethoden für die Projektierung, one ed., Verlagsgesellschaft Rudolf Müller, Köln-Braunsfeld, 1971.
[16] Kurt. Reimer, Die Bewegung der Menschenmassen in Verkehrsräumen, Glasers Ann. 71 (7) (1947) 121–131.
[17] M.I. Nazir, K.M.A. Al Razi, Quazi Sazzad Hossain, S.K. Adhilkary, Pedestrian flow characteristics at walkways in Rajshahi metropolitan city of Bangladesh, in: Proceedings ICCESD 2014: 2nd International Conference on Civil Engineering for Sustainable Development, 14-16 Februrary, 2014, Khulna, Bangladesh, ICCESD-2014, Department of Civil Engineering, Khulna University of Engineering & Technology, Khulna, Bangladesh, 2014, pp. 1166–1172.
[18] Henry K. Evans (Ed.), Traffic Engineering Handbook, second ed., Institute of Traffic Engineers, New Haven, Connecticut, 1950.
[19] Helmut Kirsch, Leistungsfähigkeit und Dimensionierung von Fussgängerüberwegen (Dissertation), Technische Hochschule Aachen, Aachen, 1962.
[20] Houman Hediyeh, Tarek Sayed, Mohamed H. Zaki, The use of gait parameters to evaluate pedestrian behavior at scramble phase signalized intersections, J. Adv. Transp. 49 (4) (2015) 523–534. http://dx.doi.org/10.1002/atr.1284.
[21] National Bureau of Standards. Design and Construction of Building Exits. National Bureau of Standards Miscellaneous Publications M151. Washington: Department of Commerce, 1935.
[22] Majed Almejmaj, Brian J. Meacham, Observations from student exercises to collect human behavior and movement data, in: Ulrich Weidmann, Uwe Kirsch, Michael Schreckenberg (Eds.), Pedestrian and Evacuation Dynamics 2012, Springer International Publishing, 2014, pp. 401–409. http://dx.doi.org/10.1007/978-3-319-02447-9_33.
[23] Winnie Daamen, Serge Hoogendoorn, Experimental research of pedestrian walking behavior, Transp. Res. Rec.: J. Transp. Res. Board 1828 (January) (2003) 20–30. http://dx.doi.org/10.3141/1828-03.
[24] Winnie Daamen, Serge P. Hoogendoorn, Mario Campanella, Dirk Versluis, Interaction behavior between individual Pedestrians, in: Ulrich Weidmann, Uwe Kirsch, Michael Schreckenberg (Eds.), Pedestrian and Evacuation Dynamics 2012, Springer International Publishing, 2014, pp. 1305–1313. http://dx.doi.org/10.1007/978-3-319-02447-9_107.

[25] Marija Nikolić, Michel Bierlaire, Bilal Farooq, Matthieu de Lapparent, Probabilistic speed–density relationship for Pedestrian traffic, Transp. Res. B 89 (July) (2016) 58–81. http://dx.doi.org/10.1016/j.trb.2016.04.002.
[26] Gerhard Scholz, Geschwindigkeit und Energieaufwand beim Gehen - ein Beitrag zur Untersuchung der Grundlagen für die zweckmässige Gestaltung von Fussgängerverkehrsanlagen (Dissertation), Technische Hochschule Hannover, Hannover, 1952.
[27] Lindsay W. Ludlow, Peter G. Weyand, Energy expenditure during level human walking: Seeking a simple and accurate predictive solution, J. Appl. Physiol. 120 (5) (2016) 481–494. http://dx.doi.org/10.1152/japplphysiol.00864.2015.
[28] Peter G. Weyand, Bethany R. Smith, Nicole S. Schultz, Lindsay W. Ludlow, Maurice R. Puyau, Nancy F. Butte., Predicting metabolic rate across walking speed: One fit for all body sizes? J. Appl. Physiol. 115 (9) (2013) 1332–1342. http://dx.doi.org/10.1152/japplphysiol.01333.2012.
[29] Raymond C. Browning, Emily A. Baker, Jessica A. Herron, Rodger Kram, Effects of obesity and sex on the energetic cost and preferred speed of walking, J. Appl. Physiol. 100 (2) (2006) 390–398. http://dx.doi.org/10.1152/japplphysiol.00767.2005.
[30] Stephanie Studenski, Subashan Perera, Kushang Patel, Caterina Rosano, Kimberly Faulkner, Marco Inzitari, Jennifer Brach, et al., Gait speed and survival in older adults, JAMA 305 (1) (2011) 50–58. http://dx.doi.org/10.1001/jama.2010.1923.
[31] Marc H. Bornstein, The pace of life: Revisited, Int. J. Psychol. 14 (1–4) (1979) 83–90. http://dx.doi.org/10.1080/00207597908246715.
[32] Robert V. Levine, Ara Norenzayan, The pace of life in 31 countries, J. Cross-Cultural Psychol. 30 (2) (1999) 178–205. http://dx.doi.org/10.1177/0022022199030002003.
[33] Robert V. Levine, The pace of life across cultures, in: Joseph Edward McGrath (Ed.), The Social Psychology of Time: New Perspectives, Sage Publications, Newbury Park, 1988, pp. 39–60.
[34] Alain Schmitt, Klauz Atzwanger, Walking fast-ranking high: A sociobiological perspective on pace, Ethol. Sociobiol. 16 (5) (1995) 451–462. http://dx.doi.org/10.1016/0162-30959500070-4.
[35] K. Daeves, O. Flachsbart, Strömung von Menschenmengen, Z. Ver. Dtsch. Ing. 94 (26) (1952) 880–881.
[36] Motoshige Isobe, Taku Adachi, Takashi Nagatani, Experiment and simulation of pedestrian counter flow, Physica A 336 (3–4) (2004) 638–650. http://dx.doi.org/10.1016/j.physa.2004.01.043.
[37] Roland Müller, Pedestrian Volume and Speed Characteristics, Yale University, Faculty of the Bureau of Highway Traffic, Yale, 1968.
[38] Detlef Oeding, Verkehrsbelastung und Dimensionierung von Gehwegen und anderen Anlagen des Fussgängerverkehrs. (Dissertation), Technische Hochschule Carolo-Wilhelmina zu Braunschweig, Braunschweig, 1963.
[39] Armin Seyfried, Bernhard Steffen, Wolfram Klingsch, Maik Boltes, The fundamental diagram of Pedestrian movement revisited, J. Stat. Mech. Theory Exp. 2005 (10) (2005) 1–13. http://dx.doi.org/10.1088/1742-5468/2005/10/P10002.
[40] Chung Yu Cheung, W.H.K. Lam, A study of the bi-directional pedestrian flow characteristics in Hong Kong mass transit railway stations, J. Eastern Asia Soc. Transp. Stud. 2 (5) (1997) 1607–1619.
[41] Houman Hediyeh, Tarek Sayed, Mohamed H. Zaki, Karim Ismail, Automated analysis of pedestrian crossing speed behavior at scramble-phase signalized intersections using computer vision techniques, Int. J. Sustain. Transp. 8 (5) (2014) 382–397. http://dx.doi.org/10.1080/15568318.2012.708098.
[42] L.F. Henderson, D.M. Jenkins, Response of Pedestrians to traffic challenge, Transp. Res. 8 (1) (1974) 71–74. http://dx.doi.org/10.1016/0041-16477490019-7.
[43] K.E. Boyce, T.J. Shields, G.W.H. Silcock, Toward the characterization of building occupancies for fire safety engineering: Capabilities of disabled people moving horizontally and on an incline, Fire Technol. 35 (1) (1999) 51–67. http://dx.doi.org/10.1023/A:1015339216366.
[44] Jake L. Pauls, Building evacuation: Research findings and recommendations, in: David Canter (Ed.), Fires and Human Behaviour, J. Wiley, Chichester, New York, 1980, pp. 251–275.
[45] R.D. Peacock, B.L. Hoskins, E.D. Kuligowski, Overall and local movement speeds during fire drill evacuations in buildings up to 31 stories, in: Richard D. Peacock, Erica D. Kuligowski, Jason D. Averill (Eds.), Pedestrian and Evacuation Dynamics, Springer, US, 2011, pp. 25–35. http://dx.doi.org/10.1007/978-1-4419-9725-8_3.
[46] Winnie. Daamen, Modelling Passenger Flows in Public Transport Facilities, in: Trail Thesis Series, Netherlands TRAIL Research School, Delft, 2004.
[47] Dirk Helbing, Péter Molnár, Social force model for pedestrian dynamics, Phys. Rev. E 51 (5) (1995) 4282–4286. http://dx.doi.org/10.1103/PhysRevE.51.4282.
[48] Serge P. Hoogendoorn, Femke van Wageningen-Kessels, Winnie Daamen, Dorine C. Duives, Continuum modelling of Pedestrian flows: From microscopic principles to self-organised macroscopic phenomena, Physica A (2014) http://dx.doi.org/10.1016/j.physa.2014.07.050, August.
[49] Gunnar G. Løvås, Modeling and simulation of pedestrian traffic flow, Transp. Res. B 28 (6) (1994) 429–443. http://dx.doi.org/10.1016/0191-26159490013-2.
[50] Daniel R. Parisi, Marcelo Gilman, Herman Moldovan, A modification of the social force model can reproduce experimental data of pedestrian flows in normal conditions, Physica A 388 (17) (2009) 3600–3608. http://dx.doi.org/10.1016/j.physa.2009.05.027.
[51] Xuan Liu, Weiguo Song, Jun Zhang, Extraction and quantitative analysis of microscopic evacuation characteristics based on digital image processing, Physica A 388 (13) (2009) 2717–2726. http://dx.doi.org/10.1016/j.physa.2009.03.017.
[52] Serge P. Hoogendoorn, W. Daamen, Pedestrian behavior at bottlenecks, Transp. Sci. 39 (2) (2005) 147–159. http://dx.doi.org/10.1287/trsc.1040.0102.
[53] Armin Seyfried, Bernhard Steffen, Thomas Lippert, Basics of modelling the Pedestrian flow, Physica A 368 (1) (2006) 232–238. http://dx.doi.org/10.1016/j.physa.2005.11.052.
[54] J. Zhang, A. Seyfried, Comparison of intersecting Pedestrian flows based on experiments, Physica A 405 (July) (2014) 316–325. http://dx.doi.org/10.1016/j.physa.2014.03.004.
[55] G.A. Dean, An analysis of the energy expenditure in level and grade walking, Ergonomics 8 (1) (1965) 31–47. http://dx.doi.org/10.1080/00140136508930772.
[56] H.J. Ralston, Energy-speed relation and optimal speed during level walking, Int. Z. Angew. Physiol. Einschließich Arbeitsphysiol. 17 (4) (1958) 277–283. http://dx.doi.org/10.1007/BF00698754.
[57] Gerhard Scholz, Die Beziehung zwischen Schrittlänge und Schrittzahl beim natürlichen Gang, Arbeitsphysiologie 15 (3) (1953) 211–222. http://dx.doi.org/10.1007/BF00933317.
[58] P.G. Weyand, B.R. Smith, M.R. Puyau, N.F. Butte, The mass-specific energy cost of human walking is set by stature, J. Exp. Biol. 213 (23) (2010) 3972–3979. http://dx.doi.org/10.1242/jeb.048199.
[59] United Nations, Department of Economic and Social Affairs, Population Division. World Population Prospects: The 2012 Revision, DVD Edition, 2013.
[60] Reto Da Forno, Michael Dahinden, Benjamin Dissler, Lennart Elsen, Lukas Gratwohl, Speed of life: Wie schnell äuft ein Schweizer? ETH Zürich, Zürich, 2011.
[61] D. Jim Walmsley, Gareth J. Lewis, The pace of Pedestrian flows in cities, Environ. Behav. 21 (2) (1989) 123–150. http://dx.doi.org/10.1177/0013916589212001.
[62] Lester A. Hoel, Pedestrian travel rates in central business districts, Traffic Eng. 39 (1968) 10–13.
[63] Rolf. Monheim, Fussgängerbereiche und Fussgängerverkehr in Stadtzentren in der Bundesrepublik Deutschland, in: Bonner Geographische Abhandlungen, vol. 64, Ferd. Dümmlers Verlag, Bonn, 1980.
[64] Department of the Army, 1990, Field Manual 21-18 Foot Marches, 21-18, Washington.
[65] Wolfgang Riepl, Das Nachrichtenwesen des Altertums: mit besonderer Rücksicht auf die Römer, Olms, Hildesheim/New York, 1972.
[66] K. Baedeker, Die Schweiz, die italienischen Seen, Mailand, Turin, Genua, Nizza. K. Baedeker, 1859.
[67] G. Buchner, Das Wissenswürdigste aus der Mass-, Gewichts- und Münz-Kunde in tabellarischen Darstellung mit bestimmter Berücksichtigung des bayerischen Mass- und Gewichts-System. Paut, 1853.
[68] Feer, Nachtrag über die wahre Grösse der Schweizerischen Stunden von 18000 Berner und 15000 Zürcher Fuss, in: Freyherrn von Zach, Achter Band (Eds.), Monatliche Correspondenz zur Beförderung der Erd- und Himmels-Kunde, Verlag der Beckerischen Buchhandlung, Gotha, 1803, pp. 318–320.
[69] Ignaz. Lindner, Logarithmisches und logarithmisch-trigonometrisches Handbuch: mit einem Anhange von mehreren für die Ausübung unentbehrlichen Tafeln und Formeln, zum Gebrauche der Mathematik Beflissenen in der k. k. österr. Artillerie. Wien: Tendler, 1831.

[70] Johann Christian Nelkenbrecher, F. Wolff, Allgemeines Taschenbuch der Mass-, Gewichts- und Münzkunde: Für Banquiers und Kaufleute. Mit neuen Münz-Tabellen versehen von H. C. Kandelhardt. Sander, 1842.
[71] Ernst Siegfried Mittler (Ed.) Zeitschrift für Kunst, Wissenschaft und Geschichte des Krieges. Mittler, 1830.
[72] Deutscher Alpenverein, Österreichischer Alpenverein, Wegehandbuch der Alpenvereine. München/Innsbruck: Deutscher und Österreichischer Alpenverein, 2011.
[73] Christian Hadorn, Schweizer Wanderwege, Signalisation Wanderwege, Bundesamt für Strassen ASTRA, Bern, 2013.
[74] Timothy J. Hatton, Bernice E. Bray, Long run trends in the heights of European men, 19th–20th centuries, Econ. Hum. Biol. 8 (3) (2010) 405–413. http://dx.doi.org/10.1016/j.ehb.2010.03.001.
[75] Dražen Brščić, Francesco Zanlungo, Takayuki Kanda, Density and velocity patterns during one year of pedestrian tracking, Transp. Res. Proc. 2 (2014) 77–86. http://dx.doi.org/10.1016/j.trpro.2014.09.011.
[76] Chalat Tipakornkiat, Limanond Thirayoot, Hyunmyung Kim, Determining an influencing area affecting walking speed on footpath: A case study of a footpath in CBD Bangkok, Thailand, Physica A 391 (22) (2012) 5453–5464. http://dx.doi.org/10.1016/j.physa.2012.06.001.